

$$f(x) = x \text{ en } -\pi \leq x \leq \pi$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) \cdot dx$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} x dx = \frac{1}{\pi} \cdot \frac{x^2}{2} \Big|_{-\pi}^{\pi} = \frac{1}{\pi} \left(\frac{\pi^2}{2} - \frac{(-\pi)^2}{2} \right) = \frac{1}{\pi} \left(\frac{\pi^2}{2} - \frac{\pi^2}{2} \right) = 0$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cdot \cos\left(\frac{n\pi x}{L}\right) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cdot \cos(nx) dx$$

$$\int u dv = uv - \int u dv$$

$$u = x \quad dv = \cos(nx) dx$$

$$du = dx \quad v = -\frac{1}{n} \sin(nx)$$

$$\int \cos(ax) = \frac{1}{a} \sin(ax)$$

$$\int \cos(ax) = -\frac{1}{a} \sin(ax)$$

$$a_n = \frac{1}{\pi} \left(\frac{1}{n} x \cdot \sin(nx) \Big|_{-\pi}^{\pi} - \frac{1}{\pi} \int_{-\pi}^{\pi} \sin(nx) dx \right)$$

$$a_n = \frac{1}{\pi} \left(\frac{1}{n} \pi \cdot \sin(n\pi) - \frac{1}{n} (-\pi) \cdot \sin(-n\pi) + \frac{1}{n^2} \cdot \cos(nx) \Big|_{-\pi}^{\pi} \right)$$

$$a_n = \frac{1}{\pi n^2} (\cos(n\pi) - \cos(-n\pi)) = 0$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cdot \sin\left(\frac{n\pi x}{L}\right) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cdot \sin(nx) dx$$

$$\int u dv = uv - \int u dv$$

$$u = x \quad dv = \sin(nx) dx$$

$$du = dx \quad v = -\frac{1}{n} \cos(nx)$$

$$\int \sin(ax) = -\frac{1}{a} \cos(ax)$$

$$\int \cos(ax) = \frac{1}{a} \sin(ax)$$

$$b_n = \frac{1}{\pi} \left(-\frac{1}{n} x \cdot \cos(nx) \Big|_{-\pi}^{\pi} - \frac{1}{\pi} \int_{-\pi}^{\pi} \cos(nx) dx \right)$$

$$b_n = \frac{1}{\pi} \left(-\frac{1}{n} \pi \cdot \cos(n\pi) - \frac{1}{n} (-\pi) \cdot \cos(-n\pi) + \frac{1}{n^2} \cdot \text{sen}(nx) \Big|_{-\pi}^{\pi} \right)$$

$$b_n = \frac{1}{\pi} \left(-\frac{2\pi}{n} \cos(n\pi) + \frac{1}{n^2} \cdot \text{sen}(n\pi) - \frac{1}{n^2} \cdot \text{sen}(-n\pi) \right) = 0$$

$$\square \cos(n\pi) = (-1)^n$$

$$b_n = -\frac{2}{n} (-1)^n = \frac{2}{n} (-1)^{n+1}$$

$$\square f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \text{sen}\left(\frac{n\pi x}{L}\right)$$

$$a_0 = 0 \quad a_n = 0 \quad b_n = \frac{2}{n} (-1)^{n+1}$$

$$x = \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \text{sen}(nx)$$

$$x = 2 \left(\text{sen}(x) - \frac{1}{2} \text{sen}(2x) + \frac{1}{3} \text{sen}(3x) - \frac{1}{4} \text{sen}(4x) + \dots \right)$$

$$f(x) = \begin{cases} 1, & -3 \leq x < 0 \\ 2, & 0 \leq x \leq 3 \end{cases}$$

$$\square f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \operatorname{sen}\left(\frac{n\pi x}{L}\right)$$

$$\square a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

$$a_0 = \frac{1}{3} \int_{-3}^0 1 dx + \frac{1}{3} \int_0^3 2 dx$$

$$a_0 = \frac{1}{3} x \Big|_{-3}^0 + \frac{1}{3} 2x \Big|_0^3 = \frac{1}{3} (0 - (-3)) + \frac{1}{3} (2(3) - 2(0))$$

$$a_0 = \frac{1}{3} (3) + \frac{1}{3} (6) = 1 + 2 = 3$$

$$\square a_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{3} \int_{-3}^0 1 \cdot \cos\left(\frac{n\pi x}{3}\right) dx + \frac{1}{3} \int_0^3 2 \cdot \cos\left(\frac{n\pi x}{3}\right) dx$$

$$a_n = \frac{1}{3} \cdot \frac{3}{n\pi} \operatorname{sen}\left(\frac{n\pi x}{3}\right) \Big|_{-3}^0 + \frac{2}{3} \cdot \frac{3}{n\pi} \operatorname{sen}\left(\frac{n\pi x}{3}\right) \Big|_0^3$$

$$a_n = \frac{1}{n\pi} \left[\operatorname{sen}(0) - \operatorname{sen}\left(\frac{-3n\pi}{3}\right) \right] + \frac{2}{n\pi} \left[\operatorname{sen}\left(\frac{3n\pi}{3}\right) - \operatorname{sen}(0) \right]$$

$$a_n = \frac{1}{n\pi} [\operatorname{sen}(0) - \operatorname{sen}(-n\pi)] + \frac{2}{n\pi} [\operatorname{sen}(n\pi) - \operatorname{sen}(0)]$$

$$a_n = 0$$

$$\square b_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \operatorname{sen}\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{1}{3} \int_{-3}^0 1 \cdot \operatorname{sen}\left(\frac{n\pi x}{3}\right) dx + \frac{1}{3} \int_0^3 2 \cdot \operatorname{sen}\left(\frac{n\pi x}{3}\right) dx$$

$$b_n = -\frac{1}{3} \cdot \frac{3}{n\pi} \cos\left(\frac{n\pi x}{3}\right) \Big|_{-3}^0 - \frac{2}{3} \cdot \frac{3}{n\pi} \cos\left(\frac{n\pi x}{3}\right) \Big|_0^3$$

$$b_n = -\frac{1}{n\pi} \left[\cos(0) - \cos\left(\frac{-3n\pi}{3}\right) \right] - \frac{2}{n\pi} \left[\cos\left(\frac{3n\pi}{3}\right) - \cos(0) \right]$$

$$b_n = -\frac{1}{n\pi} [1 - (-1)^n] - \frac{2}{n\pi} [(-1)^n - 1]$$

$$b_n = \frac{1}{n\pi} - \frac{1}{n\pi} (-1)^n = \frac{1}{n\pi} [1 - (-1)^n]$$

$$\square f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \operatorname{sen}\left(\frac{n\pi x}{L}\right) \right]$$

$$a_n = 3 \quad a_n = 0 \quad b_n = \frac{1}{n\pi} [1 - (-1)^n]$$

$$f(x) = \frac{3}{2} + \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n\pi} \operatorname{sen}\left(\frac{n\pi x}{3}\right)$$

$$f(x) = \frac{3}{2} + \frac{2}{1\pi} \operatorname{sen}\left(\frac{1\pi x}{3}\right) + \frac{2}{3\pi} \operatorname{sen}\left(\frac{3\pi x}{3}\right) + \frac{3}{5\pi} \operatorname{sen}\left(\frac{5\pi x}{3}\right) + \frac{2}{7\pi} \operatorname{sen}\left(\frac{7\pi x}{3}\right) + \dots$$

$$f(x) = \frac{3}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \operatorname{sen}\left(\frac{(2n-1)\pi x}{3}\right)$$

$$f(x) = x^2 - x + 3 \text{ en } -2 \leq x \leq 2$$

$$a_0 = \frac{1}{L} \int_L^L x dx$$

$$a_0 = \frac{1}{2} \int_{-2}^2 (x^2 - x + 3) dx = \frac{1}{2} \left(\frac{x^3}{3} - \frac{x^2}{2} + 3x \right) \Big|_{-2}^2$$

$$a_0 = \frac{1}{2} \left[\frac{(2)^3}{3} - \frac{(2)^2}{2} + 3(2) - \left(\frac{(-2)^3}{3} - \frac{(-2)^2}{2} + 3(-2) \right) \right]$$

$$a_0 = \frac{1}{2} \left[\frac{8}{3} - \frac{4}{2} + 6 - \left(-\frac{8}{3} - \frac{4}{2} + 3(-2) \right) \right] = \frac{1}{2} \left[\frac{8}{3} + 4 + \frac{8}{3} + 8 \right] = \frac{1}{2} \left[\frac{52}{3} \right] = \frac{26}{3}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{2} \int_{-2}^2 (x^2 - x + 3) \cdot \cos\left(\frac{n\pi x}{2}\right) dx$$

Método tabular

$$x^2 - x + 3 \quad \cos\left(\frac{n\pi x}{2}\right)$$

$$2x - 1 \quad \frac{2}{n\pi} \operatorname{sen}\left(\frac{n\pi x}{2}\right)$$

$$2 \quad \frac{-4}{n^2 \pi^2} \cos\left(\frac{n\pi x}{2}\right)$$

$$0 \quad \frac{-8}{n^3 \pi^3} \operatorname{sen}\left(\frac{n\pi x}{2}\right)$$

$$a_n = \frac{1}{2} (2x - 1) \left(\frac{4}{n^2 \pi^2} \cos\left(\frac{n\pi x}{2}\right) \right) \Big|_{-2}^2$$

$$a_n = \frac{2}{n^2 \pi^2} (2x - 1) \left(\cos\left(\frac{n\pi x}{2}\right) \right) \Big|_{-2}^2$$

$$a_n = \frac{2}{n^2 \pi^2} [3(\cos(n\pi)) - (-5)\cos(-n\pi)]$$

$$\cos(n\pi) = (-1)^n$$

$$\cos(-n\pi) = (-1)^n$$

$$a_n = \frac{2}{n^2 \pi^2} [3(-1)^n + 5(-1)^n] = \frac{2}{n^2 \pi^2} [8(-1)^n]$$

$$a_n = \frac{16(-1)^n}{n^2 \pi^2}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \operatorname{sen}\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{1}{2} \int_{-2}^2 (x^2 - x + 3) \cdot \operatorname{sen}\left(\frac{n\pi x}{2}\right) dx$$

$$b_n = \frac{1}{2}(2x-1) \left(\frac{4}{n^2 \pi^2} \cos\left(\frac{n\pi x}{2}\right) \right) \Big|_{-2}^2$$

Método tabular

$$x^2 - x + 3 \quad \text{sen}\left(\frac{n\pi x}{2}\right)$$

$$2x - 1 \quad \frac{-2}{n\pi} \cos\left(\frac{n\pi x}{2}\right)$$

$$2 \quad \frac{-4}{n^2 \pi^2} \text{sen}\left(\frac{n\pi x}{2}\right)$$

$$0 \quad \frac{8}{n^3 \pi^3} \cos\left(\frac{n\pi x}{2}\right)$$

$$b_n = \frac{1}{2} \left[-\frac{2}{n\pi} (x^2 - x + 3) \cos\left(\frac{n\pi x}{2}\right) + \frac{16}{n^3 \pi^3} \cos\left(\frac{n\pi x}{2}\right) \right] \Big|_{-2}^2$$

$$b_n = \frac{1}{2} \left[-\frac{2}{n\pi} (5) \cos(n\pi) + \frac{16}{n^3 \pi^3} \cos(n\pi) \right] - \frac{1}{2} \left[-\frac{2}{n\pi} (9) \cos(-n\pi) + \frac{16}{n^3 \pi^3} \cos(-n\pi) \right]$$

$$b_n = \frac{1}{2} \left[-\frac{10}{n\pi} (-1)^n + \frac{16}{n^3 \pi^3} (-1)^n \right] - \frac{1}{2} \left[-\frac{18}{n\pi} (-1)^n + \frac{16}{n^3 \pi^3} (-1)^n \right]$$

$$b_n = -\frac{5}{n\pi} (-1)^n + \frac{8}{n^3 \pi^3} (-1)^n + \frac{9}{n\pi} (-1)^n - \frac{8}{n^3 \pi^3} (-1)^n = \frac{4}{n\pi} (-1)^n$$

$$\square f(x) = \frac{a_0}{2} \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \text{sen}\left(\frac{n\pi x}{L}\right) \right]$$

$$f(x) = \frac{13}{3} + \sum_{n=1}^{\infty} \left[\frac{16(-1)^n}{n^2 \pi^2} \cos\left(\frac{n\pi x}{2}\right) + \frac{4(-1)^n}{n\pi} \text{sen}\left(\frac{n\pi x}{2}\right) \right]$$

$$f(x) = \frac{13}{3} - \frac{16}{\pi^2} \cos\left(\frac{\pi x}{2}\right) - \frac{4}{\pi} \text{sen}\left(\frac{\pi x}{2}\right) + \frac{4}{\pi^2} \cos(\pi x) - \frac{2}{\pi} \text{sen}(\pi x) \dots$$

$$f(x) = x(3-x) \text{ en } 0 \leq x \leq 3$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{2}{3} \int_0^3 (3x - x^2) \cdot \sin\left(\frac{n\pi x}{3}\right) dx$$

Método tabular

$$3x - x^2 \quad \sin\left(\frac{n\pi x}{3}\right)$$

$$3 - 2x \quad -\frac{3}{n\pi} \cos\left(\frac{n\pi x}{3}\right)$$

$$-2 \quad -\frac{9}{n^2 \pi^2} \sin\left(\frac{n\pi x}{3}\right)$$

$$0 \quad \frac{27}{n^3 \pi^3} \cos\left(\frac{n\pi x}{3}\right)$$

$$b_n = \frac{2}{3} \left(-\frac{3}{n\pi} (3x - x^2) \cos\left(\frac{n\pi x}{3}\right) \Big|_0^3 + \frac{9}{n^2 \pi^2} (3 - 2x) \sin\left(\frac{n\pi x}{3}\right) \Big|_0^3 - \frac{54}{n^3 \pi^3} \cos\left(\frac{n\pi x}{3}\right) \Big|_0^3 \right)$$

$$b_n = \frac{2}{3} \left(-\frac{3}{n\pi} (9 - 9) \cos(n\pi) + \frac{3}{n\pi} (0 - 0) \cos(0) + \frac{9}{n^2 \pi^2} (3 - 6) \sin(n\pi) \right. \\ \left. - \frac{9}{n^2 \pi^2} (3 - 2x) \sin\left(\frac{n\pi x}{3}\right) \Big|_0^3 - \frac{54}{n^3 \pi^3} \cos(n\pi) + \frac{54}{n^3 \pi^3} \cos(0) \right)$$

$$b_n = \frac{2}{3} \left(-\frac{54}{n^3 \pi^3} (-1)^n + \frac{54}{n^3 \pi^3} \right) = \frac{2}{3} \left(\frac{54}{n^3 \pi^3} (1 - (-1)^n) \right)$$

$$b_n = \frac{36}{n^3 \pi^3} (1 - (-1)^n)$$

si $n = 2k$

$$b_{2k} = \frac{36}{(2k)^3 \pi^3} (1 - (-1)^{2k}) = \frac{36}{(2k)^3 \pi^3} (1 - 1) = 0$$

si $n = 2k + 1$

$$b_{2k+1} = \frac{36}{(2k+1)^3 \pi^3} (1 - (-1)^{2k+1}) = \frac{36}{(2k+1)^3 \pi^3} (1 - (-1)) = \frac{72}{(2k+1)^3 \pi^3}$$

$$\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{3}\right) = \sum_{n=0}^{\infty} b_{2k+1} \sin\left(\frac{(2k+1)\pi x}{3}\right)$$

$$\sum_{n=0}^{\infty} \frac{72}{(2k+1)^3 \pi^3} \sin\left(\frac{(2k+1)\pi x}{3}\right) = \sum_{n=0}^{\infty} \frac{72}{(2n+1)^3 \pi^3} \sin\left(\frac{(2n+1)\pi x}{3}\right)$$

$$f(x) = x^2(2-x) \text{ en } 0 \leq x \leq 2$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \text{sen}\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{2}{2} \int_0^2 (3x - x^2) \cdot \text{sen}\left(\frac{n\pi x}{2}\right) dx$$

Método tabular

$$2x^2 - x^3 \quad \text{sen}\left(\frac{n\pi x}{2}\right)$$

$$4x - 3x^2 \quad -\frac{2}{n\pi} \cos\left(\frac{n\pi x}{2}\right)$$

$$4 - 6x \quad -\frac{4}{n^2\pi^2} \text{sen}\left(\frac{n\pi x}{2}\right)$$

$$-6 \quad \frac{8}{n^3\pi^3} \cos\left(\frac{n\pi x}{2}\right)$$

$$0 \quad \frac{16}{n^4\pi^4} \text{sen}\left(\frac{n\pi x}{2}\right)$$

$$b_n = \frac{2}{2} \left(-\frac{2}{n\pi} (2x^2 - x^3) \cos\left(\frac{n\pi x}{2}\right) \Big|_0^2 + \frac{4}{n^2\pi^2} (4x - 3x^2) \text{sen}\left(\frac{n\pi x}{2}\right) \Big|_0^2 \right. \\ \left. + \frac{8}{n^3\pi^3} (4 - 6x) \cos\left(\frac{n\pi x}{2}\right) \Big|_0^2 + \frac{96}{n^4\pi^4} \text{sen}\left(\frac{n\pi x}{2}\right) \Big|_0^2 \right)$$

$$b_n = -\frac{2}{n\pi} (2x^2 - x^3) \cos\left(\frac{n\pi x}{2}\right) \Big|_0^2 + \frac{4}{n^2\pi^2} (4x - 3x^2) \text{sen}\left(\frac{n\pi x}{2}\right) \Big|_0^2$$

$$+ \frac{8}{n^3\pi^3} (4 - 6x) \cos\left(\frac{n\pi x}{2}\right) \Big|_0^2 + \frac{96}{n^4\pi^4} \text{sen}\left(\frac{n\pi x}{2}\right) \Big|_0^2$$

$$b_n = -\frac{2}{n\pi} (8 - 8) \cos(n\pi) + \frac{2}{n\pi} (0) \cos(0) + \frac{4}{n^2\pi^2} (8 - 12) \text{sen}(n\pi) - \frac{4}{n^2\pi^2} (0) \text{sen}(0)$$

$$+ \frac{8}{n^3\pi^3} (4 - 12) \cos(n\pi) - \frac{8}{n^3\pi^3} (4) \cos(0) + \frac{96}{n^4\pi^4} \text{sen}(n\pi) - \frac{96}{n^4\pi^4} \text{sen}(0)$$

$$b_n = -\frac{64}{n^3\pi^3} (-1)^n - \frac{32}{n^3\pi^3}$$

$$b_n = -\frac{32}{n^3\pi^3} [2(-1)^n + 1]$$

$$x^2(2-x) = \sum_{n=1}^{\infty} \left(-\frac{32}{n^3\pi^3} \right) [2(-1)^n + 1] \text{sen}\left(\frac{n\pi x}{2}\right)$$

$$x^2(2-x) = -\frac{32}{\pi^3} \sum_{n=1}^{\infty} \frac{[2(-1)^n + 1]}{n^3} \text{sen}\left(\frac{n\pi x}{2}\right)$$

$$f(t) = \begin{cases} 2t, & 0 < t < 1 \\ -2t + 4, & 1 < t < 2 \end{cases}$$

$$T = 4$$

Función IMPAR

$$a_0 = 0; a_n = 0$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \cdot \sin\left(\frac{n\pi t}{L}\right) dt$$

$$b_n = \frac{2}{2} \int_0^2 f(t) \cdot \sin\left(\frac{n\pi t}{2}\right) dt = \int_0^1 2t \cdot \sin\left(\frac{n\pi t}{2}\right) dt + \int_1^2 (-2t + 4) \sin\left(\frac{n\pi t}{2}\right) dt$$

Integración por partes

$$\int u dv = uv - \int v du$$

$$u = 2t \quad dv = \sin\left(\frac{n\pi t}{2}\right) dt$$

$$du = 2dt \quad v = -\frac{2}{n\pi} \cos\left(\frac{n\pi t}{2}\right)$$

Integración por partes

$$\int u dv = uv - \int v du$$

$$u = -2t + 4 \quad dv = \sin\left(\frac{n\pi t}{2}\right) dt$$

$$du = -2dt \quad v = -\frac{2}{n\pi} \cos\left(\frac{n\pi t}{2}\right)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \sin\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = -\frac{2}{n\pi} \left(2t \cos\left(\frac{n\pi t}{2}\right) \right)_0^1 - \int_0^1 \left(-\frac{2}{n\pi} \right) \cos\left(\frac{n\pi t}{2}\right) (2dt) + (-2t + 4) \left(-\frac{2}{n\pi} \right) \cos\left(\frac{n\pi t}{2}\right) \Big|_1^2 - \int_1^2 \left(-\frac{2}{n\pi} \right) \cos\left(\frac{n\pi t}{2}\right) (-2dt)$$

$$b_n = -\frac{4}{n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{4}{n\pi} \int_0^1 \cos\left(\frac{n\pi t}{2}\right) dt + \frac{4}{n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{4}{n\pi} \int_1^2 \cos\left(\frac{n\pi t}{2}\right) dt$$

$$b_n = -\frac{4}{n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{8}{n^2 \pi^2} \sin\left(\frac{n\pi t}{2}\right) \Big|_0^1 + \frac{4}{n\pi} \cos\left(\frac{n\pi}{2}\right) - \frac{8}{n^2 \pi^2} \sin\left(\frac{n\pi t}{2}\right) \Big|_1^2$$

$$b_n = \frac{16}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right)$$

$$a_0 = 0; a_n = 0$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

$$f(t) = \sum_{n=1}^{\infty} \left[\frac{16}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi t}{2}\right) \right]$$

$$n = 2k \rightarrow 0$$

$$n = 2k + 1 \rightarrow (-1)^k \text{ Desde } = 0$$

$$f(t) = \sum_{n=1}^{\infty} \frac{16(-1)^k}{(2k+1)^2 \pi^2} \sin\left(\frac{(2k+1)\pi t}{2}\right)$$

$$f(t) = \begin{cases} -t, & -1 < t < 0 \\ 0, & 0 < t < 1 \end{cases}$$

$$T = 2$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t).dt$$

$$a_0 = \frac{1}{1} \int_{-1}^1 f(t).dt = \int_{-1}^1 -t.dt = -\frac{t^2}{2} \Big|_{-1}^1 = -\frac{0^2}{2} + \frac{(-1)^2}{2} = \frac{1}{2}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x). \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \int_{-1}^0 -t. \cos(n\pi t) dt$$

Integración por partes

$$\int u dv = uv - \int u dv$$

$$u = -t \quad dv = \cos(n\pi t) dt$$

$$du = -dt \quad v = \frac{1}{n\pi} \operatorname{sen}(n\pi t)$$

$$a_n = \frac{-t}{n\pi} \operatorname{sen}(n\pi t) \Big|_{-1}^0 + \frac{1}{n\pi} \int_{-1}^0 \operatorname{sen}(n\pi t) dt$$

$$a_n = \frac{-0}{n\pi} \operatorname{sen}(0) - \frac{-(-1)}{n\pi} \operatorname{sen}(-n\pi) - \frac{1}{n^2 \pi^2} \cos(n\pi t) \Big|_{-1}^0$$

$$a_n = -\frac{1}{n^2 \pi^2} \cos(n\pi t) \Big|_{-1}^0 = -\frac{1}{n^2 \pi^2} \cos(0) + \frac{1}{n^2 \pi^2} \cos(-n\pi)$$

$$a_n = -\frac{1}{n^2 \pi^2} + \frac{1}{n^2 \pi^2} (-1)^n$$

$$a_n = \frac{1}{n^2 \pi^2} ((-1)^n - 1)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t). \operatorname{sen}\left(\frac{n\pi t}{L}\right) dt$$

$$b_n = \int_{-1}^0 -t. \operatorname{sen}(n\pi t) dt$$

Integración por partes

$$\square \int u dv = uv - \int u dv$$

$$u = -t \quad dv = \operatorname{sen}(n\pi t) dt$$

$$du = -dt \quad v = -\frac{1}{n\pi} \cos(n\pi t)$$

$$b_n = \frac{t}{n\pi} \cos(n\pi t) \Big|_{-1}^0 - \frac{1}{n\pi} \int_{-1}^0 \cos(n\pi t) dt$$

$$b_n = \frac{0}{n\pi} \cos(0) - \frac{-1}{n\pi} \cos(-n\pi) - \frac{1}{n^2 \pi^2} \operatorname{sen}(n\pi t) \Big|_{-1}^0$$

$$b_n = \frac{1}{n\pi} (-1)^n - \frac{1}{n^2 \pi^2} \operatorname{sen}(0) + \frac{1}{n^2 \pi^2} \operatorname{sen}(-n\pi)$$

$$b_n = \frac{1}{n\pi} (-1)^n$$

$$\square f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \operatorname{sen}\left(\frac{n\pi x}{L}\right) \right]$$

$$f(t) = \frac{1}{4} \sum_{n=1}^{\infty} \left[\frac{1}{n^2 \pi^2} \left((-1)^n - 1 \right) \cos(n\pi t) + \frac{1}{n\pi} (-1)^n \operatorname{sen}(n\pi t) \right]$$

$$f(t) = \begin{cases} -t, & -2 \leq t < 0 \\ t, & 0 \leq t \leq 2 \end{cases}$$

$$L = 2$$

Función par $b_n = 0$

$$a_0 = \frac{2}{L} \int_{-L}^L f(t) dt$$

$$a_0 = \frac{2}{2} \int_0^2 t dt = \frac{t^2}{2} \Big|_0^2 = \frac{2^2}{2} - \frac{0^2}{2} = 2$$

$$a_n = \frac{2}{L} \int_{-L}^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{2}{2} \int_{-1}^0 t \cdot \cos\left(\frac{n\pi t}{2}\right) dt$$

Integración por partes

$$\int u dv = uv - \int u dv$$

$$u = t \quad dv = \cos\left(\frac{n\pi t}{2}\right) dt$$

$$du = dt \quad v = \frac{2}{n\pi} \operatorname{sen}\left(\frac{n\pi t}{2}\right)$$

$$a_n = \frac{2}{n\pi} t \operatorname{sen}\left(\frac{n\pi t}{2}\right) \Big|_0^2 - \int_0^2 \frac{2}{n\pi} \operatorname{sen}\left(\frac{n\pi t}{2}\right) dt$$

$$a_n = -\frac{2}{n\pi} \int_0^2 \frac{2}{n\pi} \operatorname{sen}\left(\frac{n\pi t}{2}\right) = -\frac{2}{n\pi} \left(-\frac{2}{n\pi} \cos\left(\frac{n\pi t}{2}\right) \Big|_0^2 \right) = \frac{4}{n^2 \pi^2} \left(\cos\left(\frac{n\pi t}{2}\right) \Big|_0^2 \right)$$

$$a_n = \frac{4}{n^2 \pi^2} \left(\cos\left(\frac{n\pi 2}{2}\right) - \cos\left(\frac{n\pi(0)}{2}\right) \right) = \frac{4}{n^2 \pi^2} (\cos(n\pi) - \cos(0))$$

$$a_n = \frac{4}{n^2 \pi^2} ((-1)^n - 1)$$

$$a_n = \frac{-0}{n\pi} \operatorname{sen}(0) - \frac{-(-1)}{n\pi} \operatorname{sen}(-n\pi) - \frac{1}{n^2 \pi^2} \cos(n\pi) \Big|_{-1}^0$$

$$a_n = -\frac{1}{n^2 \pi^2} \cos(n\pi) \Big|_{-1}^0 = -\frac{1}{n^2 \pi^2} \cos(0) + \frac{1}{n^2 \pi^2} \cos(-n\pi)$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \operatorname{sen}\left(\frac{n\pi x}{L}\right) \right]$$

$$f(t) = 1 + \sum_{n=1}^{\infty} \left[\frac{4}{n^2 \pi^2} ((-1)^n - 1) \cos\left(\frac{n\pi t}{2}\right) \right]$$

$$f(t) = \begin{cases} 1, & -1 < x < 0 \\ x, & 0 < x < 1 \end{cases}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x).dx$$

$$a_0 = \frac{1}{1} \int_{-1}^0 dx + \int_0^1 x dx = x \Big|_{-1}^0 + \frac{x^2}{2} \Big|_0^1$$

$$a_0 = (0 - (-1)) + \left(\frac{1}{2} - 0 \right) = \frac{3}{2}$$

$$a_n = \frac{2}{L} \int_{-L}^L f(x). \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{\sin(n\pi x)}{n\pi} \Big|_{-1}^0 + \frac{n\pi \cos(n\pi x) + \sin(n\pi x)}{n^2 \pi^2} \Big|_0^1$$

$$a_n = \frac{-\sin(-n\pi)}{n\pi} + \frac{n\pi \sin(n\pi) + \cos(n\pi) - 1}{n^2 \pi^2}$$

$$a_n = \frac{\cos(n\pi) - 1}{n^2 \pi^2} = \frac{(-1)^n - 1}{n^2 \pi^2}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t). \sin\left(\frac{n\pi t}{L}\right) dt$$

$$b_n = \int_{-1}^0 \sin(n\pi x) dx + \int_0^1 \sin(n\pi x) x dx$$

$$b_n = \frac{-\cos x(n\pi x)}{n\pi} \Big|_{-1}^0 + \sin(n\pi x) - n\pi \cos(n\pi x) \Big|_0^1$$

$$b_n = \frac{-1}{n\pi}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

$$f(t) = \frac{3}{4} \sum_{n=1}^{\infty} \left[\frac{(-1)^n - 1}{n^2 \pi^2} \cdot \cos(n\pi x) + \left(\frac{-1}{n\pi} \right) \cdot \sin(n\pi x) \right]$$

$$f(t) = \begin{cases} \pi + x, & -\pi \leq x \leq 0 \\ x, & 0 \leq x \leq \pi \end{cases}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x).dx$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^0 \pi + x dx + \int_0^{\pi} x dx$$

$$a_0 = \frac{1}{\pi} \left[\left(\pi^2 + \frac{x^2}{2} \right)_{-\pi}^0 + \left(\frac{x^2}{2} \right)_0^{\pi} \right]$$

$$a_0 = \frac{1}{\pi} \left[\left(\pi^2 - \frac{\pi^2}{2} \right) + \left(\frac{\pi^2}{2} \right) \right] = \frac{1}{\pi} \left[\pi^2 - \frac{\pi^2}{2} + \frac{\pi^2}{2} \right]$$

$$a_0 = \frac{1}{\pi} [\pi^2] = \pi$$

$$a_n = \frac{2}{L} \int_{-L}^L f(x). \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{x}{n} \sin(n\pi) - \frac{1}{n} (-\cos(nx))$$

$$a_n = \frac{x}{n} \sin(n\pi) + \frac{1}{n} (\cos(nx)) = 0$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t). \sin\left(\frac{n\pi t}{L}\right) dt$$

$$b_n = \frac{1}{\pi} \left[-\cos(n\pi) \pi - x. \frac{\cos(n\pi)}{n} + \frac{\cos(n\pi)}{n^2} \right]_{-\pi}^0 + \frac{1}{\pi} \left[-x. \frac{\cos(\pi)}{n} + \frac{\sin(x)}{n^2} \right]_0^{\pi}$$

$$b_n = \cos(n\pi) - 1$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(t) = \frac{\pi}{2} \sum_{n=1}^{\infty} 0(\cos(nx)) + \sum_{n=1}^{\infty} (\cos(nx) - 1). \sin(\pi x)$$

$$f(t) = \begin{cases} -x, & -2 < x < 0 \\ \frac{1}{2}, & 0 < x < 2 \end{cases}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

$$a_0 = \frac{1}{2} \int_{-2}^0 -x dx + \frac{1}{2} \int_0^2 x dx$$

$$a_0 = \frac{1}{2} \left(\frac{x^2}{2} \right)_{-2}^0 + \frac{1}{2} (x)_0^2$$

$$a_0 = \frac{1}{2} \left[\frac{(-2)^2}{2} + 1 \right] = \frac{1}{2} [3] = \frac{3}{2}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{2} \left[\int_{-2}^0 \cos\left(\frac{n\pi x}{2}\right) (-x) dx + \int_0^2 \cos\left(\frac{n\pi x}{2}\right) \left(\frac{1}{2}\right) dx \right]$$

$$a_n = 2 \cdot \frac{[(-1)^n - 1]}{n^2 x^2}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \cdot \sin\left(\frac{n\pi t}{L}\right) dt$$

$$b_n = \frac{1}{2} \left[\int_{-2}^0 \sin\left(\frac{n\pi x}{2}\right) (-x) dx + \int_0^2 \sin\left(\frac{n\pi x}{2}\right) \left(\frac{1}{2}\right) dx \right]$$

$$b_n = \frac{1}{2} \left[\frac{4(-1)^n}{\pi n} + \frac{1 - (-1)^n}{\pi} \right]$$

$$b_n = \frac{1}{2} \left[\frac{3(-1)^n}{\pi n} + 1 \right]$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(t) = \frac{3}{4} \sum_{n=1}^{\infty} 2 \cdot \left[\frac{(-1)^n - 1}{n^2 x^2} \right] \cos\left(\frac{n\pi x}{2}\right) + \frac{1}{2} \left[\frac{3(-1)^n}{\pi n} + 1 \right] \sin\left(\frac{n\pi x}{2}\right)$$

$$f(x) = \frac{2}{3}(x-1) \quad 0 < x < 4$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

$$a_0 = \frac{1}{4} \int_0^4 \frac{2}{3} x dx - \int_0^4 1 dx$$

$$a_0 = \frac{1}{4} \frac{2}{3} \int_0^4 (x-1) dx$$

$$a_0 = \frac{1}{4} \left(\left[\frac{x^2}{3} \right]_0^4 \cdot [x]_0^4 \right)$$

$$a_0 = \frac{1}{6} (8-4) = \frac{2}{3}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{4} \int_0^4 \frac{2}{3} (x-1) \cos\left(\frac{n\pi x}{4}\right) dx$$

$$a_n = \frac{1}{6} \int_0^4 x \cos\left(\frac{n\pi x}{4}\right) dx - \int_0^4 \cos\left(\frac{n\pi x}{4}\right) dx$$

$$a_n = \frac{1}{6} \left(\left[\frac{4x}{\pi n} \sin\left(\frac{n\pi x}{4}\right) + \frac{4x}{(\pi n)^2} \cos\left(\frac{n\pi x}{4}\right) \right]_0^4 - \left[\frac{4}{\pi n} \sin\left(\frac{n\pi x}{4}\right) \right]_0^4 \right)$$

$$a_n = \frac{1}{6} \left(\frac{16}{\pi^2 n^2} \cos(\pi n) - \frac{16}{(\pi n)^2} \right)$$

$$a_n = \frac{8(-1)^n}{3(\pi^2 n^2)} - \frac{8}{3(\pi^2 n^2)}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \cdot \sin\left(\frac{n\pi t}{L}\right) dt$$

$$b_n = \frac{1}{4} \int_0^4 \frac{2}{3} (x-1) \sin\left(\frac{n\pi x}{4}\right) dx$$

$$b_n = \frac{1}{6} \int_0^4 x \sin\left(\frac{n\pi x}{4}\right) dx - \int_0^4 \sin\left(\frac{n\pi x}{4}\right) dx$$

$$b_n = \frac{1}{6} \left(-\frac{4x}{\pi n} \cos\left(\frac{n\pi x}{4}\right) - \frac{16}{(\pi n)^2} \sin\left(\frac{n\pi x}{4}\right) \right)_0^4 + \frac{4x}{\pi n} \cos\left(\frac{n\pi x}{4}\right)_0^4$$

$$b_n = \frac{1}{6} \left(-\frac{16}{\pi n} (-1)^n + \frac{4}{(\pi n)^2} \sin\left(\frac{n\pi x}{4}\right) \right)_0^4 + \frac{4x}{\pi n} \cos\left(\frac{n\pi x}{4}\right)_0^4$$

$$b_n = \frac{1}{6} \left(-\frac{16}{\pi n} (-1)^n + \frac{4}{(\pi n)^2} (-1)^n - \frac{4x}{\pi n} \right)$$

$$b_n = \frac{8(-1)^{n+1}}{3\pi n} + \frac{8(-1)^n}{3\pi n} - \frac{4}{3\pi n}$$

$$b_n = \frac{(-1)^n (2-8) - 4}{3\pi n}$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(t) = \frac{2}{6} \sum_{n=1}^{\infty} \frac{8}{3(\pi^2 n^2)} \left((-1)^n - 1 \right) \cos\left(\frac{n\pi x}{4}\right) + \sum_{n=1}^{\infty} \frac{(-1)^n (2-8) - 4}{3\pi n} \sin\left(\frac{n\pi x}{4}\right)$$

$$f(t) = \begin{cases} 0, -\pi < x < 0 \\ \text{sen}x, 0 < x < \pi \end{cases}$$

$$\boxed{a_0} = \frac{1}{L} \int_{-L}^L f(x).dx$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^0 0 dx + \frac{1}{\pi} \int_0^{\pi} \text{sen}x. dx$$

$$a_0 = \frac{1}{\pi} (-\cos x)_0^{\pi}$$

$$a_0 = \frac{1}{\pi} [-\cos \pi - (-\cos(0))]]$$

$$a_0 = \frac{1}{\pi} (1+1) + \frac{2}{\pi}$$

$$\boxed{a_n} = \frac{1}{L} \int_{-L}^L f(x). \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^0 0. \cos\left(\frac{n\pi x}{\pi}\right) dx + \frac{1}{\pi} \int_0^{\pi} \text{sen}x. \cos\left(\frac{n\pi x}{\pi}\right) dx$$

$$a_n = \frac{1}{2\pi} \int_0^{\pi} 2 \text{sen}x. \cos(\pi x) dx$$

$$a_n = \frac{1}{2\pi} \int_0^{\pi} \text{sen}(n+1)x - \text{sen}(n-1)x dx$$

$$a_n = \frac{1}{2\pi} \left(\frac{-\cos(n+1)}{n+1} + \frac{\cos(n-1)}{n-1} x \right)_0^{\pi}$$

$$a_n = \frac{1}{2\pi} \left(\frac{\cos(n-1)\pi}{n-1} + \frac{\cos(0)}{n-1} - \frac{\cos(n+1)x}{n+1} + \frac{\cos(0)}{n+1} \right)$$

$$a_n = \frac{1}{2\pi} \left(\frac{(-1)^{n-1} - 1}{n-1} + \frac{1 - (-1)^{n-1}}{n+1} \right)$$

$$a_n = \frac{1}{2\pi} \left(-\frac{2}{n-1} + \frac{2}{n+1} \right)$$

$$a_n = \frac{1}{2\pi} \left(\frac{-2n-2+2n-2}{n^2-1^2} \right) = \frac{-2}{\pi(n^2-1^2)}$$

nes par 0, nes impar

$$\boxed{b_n} = \frac{1}{L} \int_{-L}^L f(t). \text{sen}\left(\frac{n\pi t}{L}\right) dt$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^0 0. \text{sen}(x) dx + \frac{1}{\pi} \int_0^{\pi} \text{sen}x. \text{sen}\left(\frac{n\pi}{\pi}\right) dx$$

$$b_n = \frac{1}{2\pi} \int_{-\pi}^0 2 \cdot \text{sen}(x) \cdot \text{sen}(x) dx$$

$$b_n = \frac{1}{2\pi} \int_{-\pi}^0 \cos(n-1)x - \cos(n+1)x dx$$

$$b_n = \frac{1}{2\pi} \left[\frac{\text{sen}(n-1)x}{n-1} - \frac{\text{sen}(n+1)x}{n+1} \right]_0^\pi$$

$$b_n = \frac{1}{2\pi} (0 - 0 + 0) = 0$$

$$\square f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \text{sen}\left(\frac{n\pi x}{L}\right)$$

$$f(t) = \frac{2\pi}{2} \sum_{n=1}^{\infty} \left[\frac{2}{\pi(n^2-1)} \cdot \frac{\cos(n\pi x)}{\pi} + 0 \cdot \frac{\text{sen}(n\pi x)}{\pi} \right]$$

$$f(t) = \begin{cases} x, & 0 < x < \pi \\ 0, & \pi < x < 2\pi \end{cases}$$

$$\square a_0 = \frac{1}{L} \int_{-L}^L f(x) \cdot dx$$

$$a_0 = \frac{1}{\pi} \left(\int_0^{\pi} x dx + \int_{\pi}^{2\pi} 0 dx \right) = \frac{1}{\pi} \left(\frac{x^2}{2} \right)_0^{\pi} = \frac{1}{\pi} \left(\frac{\pi^2}{2} \right) = \frac{\pi}{2}$$

$$\square a_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{\pi} \left(\int_0^{\pi} x \cos\left(\frac{nx}{2}\right) dx + \int_{\pi}^{2\pi} 0 \cos\left(\frac{nx}{2}\right) dx \right)$$

$$a_n = \frac{1}{\pi} \left(\int_0^{\pi} x \cos\left(\frac{nx}{2}\right) dx \right) = \frac{1}{\pi} \left(\int_0^{\frac{\pi}{2}} \frac{2u}{n} \cdot \cos(u) \cdot \frac{2u}{n} du \right)$$

$$a_n = \frac{4}{n^2 \pi} \int_0^{\frac{\pi}{2}} 4 \cos(u) du = \frac{4}{n^2 \pi} [4 \sin(u) + \cos(u)]_0^{\frac{\pi}{2}}$$

$$a_n = \frac{4}{n^2 \pi} \left(\frac{n\pi}{2} \sin\left(\frac{\pi n}{2}\right) + \cos\left(\frac{\pi n}{2}\right) - 0 \sin(0) - \cos(0) \right) = \frac{4}{n^2 \pi} \left(\frac{n\pi}{2} \sin\left(\frac{\pi n}{2}\right) + \cos\left(\frac{\pi n}{2}\right) - 1 \right)$$

$$b_n = \int_{-1}^0 -t \cdot \sin(n\pi t) dt$$

$$b_n = \frac{1}{\pi} \left(\int_0^{\pi} x \sin\left(\frac{nx}{2}\right) dx + \int_{\pi}^{2\pi} 0 \sin\left(\frac{nx}{2}\right) dx \right)$$

Integración por partes

$$\square \int u dv = uv - \int u dv$$

$$u = \frac{n}{2} \quad dv = \frac{\pi}{2} dx$$

$$du = \frac{2u}{n} \quad dx = \frac{\pi}{n} du$$

$$b_n = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} \frac{2u}{n} \sin(n) \frac{2du}{n} = \frac{4}{n^2 \pi} [u \sin(u) du]$$

$$b_n = \frac{4}{n^2 \pi} (-u \cos(u) + \sin(u))_0^{\frac{\pi}{2}}$$

$$b_n = \frac{4}{n^2 \pi} \left(-\frac{n\pi}{2} \cos\left(\frac{n\pi}{2}\right) + \sin\left(\frac{n\pi}{2}\right) \right)$$

$$\square f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(t) = \frac{\pi}{4} \sum_{n=1}^{\infty} \left[\frac{4}{n^2 \pi} \left(\frac{n\pi}{2} \sin\left(\frac{\pi n}{2}\right) + \cos\left(\frac{\pi n}{2}\right) - 1 \right) \cdot \frac{\cos(n\pi x)}{\pi} \right. \\ \left. + \frac{4}{n^2 \pi} \left(-\frac{n\pi}{2} \cos\left(\frac{n\pi}{2}\right) + \sin\left(\frac{n\pi}{2}\right) \right) \cdot \frac{\sin(n\pi x)}{\pi} \right]$$

$$f(t) = \begin{cases} 0, -1 \leq x \leq 0 \\ 2, 0 \leq x \leq 1 \end{cases}$$

$$\square a_0 = \frac{1}{L} \int_{-L}^L f(x).dx$$

$$a_0 = \frac{1}{1} \int_{-1}^0 0 dx + \frac{1}{1} \int_0^1 2 dx$$

$$a_0 = (2x)_0^1 = (2(1) - 2(0)) = 2$$

$$\square a_n = \frac{1}{L} \int_{-L}^L f(x). \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{1} \int_{-1}^0 0. \cos\left(\frac{n\pi x}{1}\right) dx + \frac{1}{1} \int_0^1 2. \cos\left(\frac{n\pi x}{1}\right) dx$$

$$a_n = 1 \int_0^1 2. \cos(n\pi x) dx$$

$$a_n = \frac{2}{n\pi} \text{sen}(n\pi x)_0^1$$

$$a_n = \frac{2}{n\pi} [\text{sen}(n\pi) - \text{sen}(0)] = 0$$

$$\square b_n = \frac{1}{L} \int_{-L}^L f(t). \text{sen}\left(\frac{n\pi t}{L}\right) dt$$

$$b_n = \frac{1}{1} \int_{-1}^0 0. \text{sen}\left(\frac{n\pi x}{1}\right) dx + \frac{1}{1} \int_0^1 -2. -\text{sen}\left(\frac{n\pi x}{1}\right) dx$$

$$b_n = 2 \int_0^1 \text{sen}(n\pi x) dx$$

$$b_n = 2 \left[-\frac{1}{\pi} \cos(n\pi x) \right]_0^1 = 0$$

$$b_n = \frac{2}{n\pi} [-\cos(n\pi) + \cos(0)]$$

$$b_n = \frac{2}{n\pi} [1 - \cos(n\pi)] = \frac{2}{n\pi} [1 - (-1)^n]$$

$$\square f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \text{sen}\left(\frac{n\pi x}{L}\right)$$

$$f(t) = 1 + \sum_{n=1}^{\infty} 0 \left(\cos\left(\frac{n\pi x}{1}\right) \right) + \frac{2}{n\pi} (1 - (-1)^n) . \text{sen}\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{x^2}{2} \quad \text{en } -\pi \leq x \leq \pi$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) \cdot dx$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{x^2}{2} dx = \frac{1}{4\pi} \int_{-\pi}^{\pi} x^2 dx$$

$$a_0 = \frac{1}{2\pi} \left(\frac{x^3}{3} \right)_{-\pi}^{\pi} = \frac{1}{4\pi} \left(\frac{\pi^3}{3} - \frac{-\pi^3}{3} \right)$$

$$a_0 = \frac{1}{4\pi} \cdot \frac{2\pi^3}{3} = \frac{\pi^2}{6}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{x^2}{2} \cdot \cos\left(\frac{n\pi x}{\pi}\right) dx = \frac{1}{4\pi} \int_{-\pi}^{\pi} x^2 \cdot \cos(nx) dx$$

Integral por partes

$$\int u dv = uv - \int u dv$$

$$u = x^2 \quad dv = \cos(nx) dx$$

$$du = 2x dx \quad v = \frac{\text{sen}(nx)}{n}$$

$$a_n = \frac{1}{4\pi} x^2 \cdot \frac{\text{sen}(nx)}{n} - \int_{-\pi}^{\pi} \frac{\text{sen}(nx)}{n} \cdot 2x dx$$

$$a_n = \frac{1}{4\pi} \left[x^2 \cdot \frac{\text{sen}(nx)}{n} \right]_{-\pi}^{\pi} - \frac{2}{n} \int_{-\pi}^{\pi} \text{sen}(nx) dx$$

Integral por partes

$$\int u dv = uv - \int u dv$$

$$u = x \quad dv = \text{sen}(nx) dx$$

$$du = dx \quad v = -\frac{\cos(nx)}{n}$$

$$a_n = \frac{1}{4\pi} \left[x^2 \cdot \frac{\text{sen}(nx)}{n} \right]_{-\pi}^{\pi} - \left[x \cdot \frac{\cos(nx)}{n} \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \frac{\cos(nx)}{n} dx$$

$$a_n = \frac{1}{4\pi} \left[\frac{x^2 \cdot \text{sen}(nx)}{n} \right]_{-\pi}^{\pi} + \frac{2}{n^2} [x \cdot \cos(nx)]_{-\pi}^{\pi} + \frac{1}{n} \int_{-\pi}^{\pi} \cos(nx) dx$$

$$a_n = \frac{1}{4\pi} \cdot \frac{2}{n^2} [\pi - \cos(\pi)] + 0$$

$$a_n = \frac{2\pi \cdot \cos(\pi)}{4\pi n^2} = \frac{\cos(\pi)}{2n^2}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \text{sen}\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{\pi^2}{12} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{\cos(\pi) \cdot \cos(n\pi)}{n^2}$$

$$f(t) = \begin{cases} 1, & -\pi \leq t \leq 0 \\ 0, & 0 \leq t \leq \pi \end{cases}$$

$$\square \frac{1}{2} a_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$\frac{1}{2} a_0 = \frac{1}{2\pi} \left[\int_{-\pi}^0 0 dt + \int_0^{\pi} 2 dt \right] = \frac{1}{2\pi} \left[(t)_{-\pi}^0 + 0 \right]$$

$$a_0 = \frac{1}{2\pi} [0 - (-\pi) + 0] = \frac{1}{2}$$

$$\square a_n = \frac{2}{T} \int_0^T f(t) \cdot \cos(n\omega_0 t) dt$$

$$a_n = \frac{2}{2\pi} \left[\int_{-1}^0 1 \cdot \cos(nt) dT + 0 \right]$$

$$\square \int \cos(u) du = \sin(u)$$

$$u = nt \quad du = n dt$$

$$\frac{1}{n} \int \cos(nt)(n dt) = \frac{1}{n} \sin(nt)$$

$$a_n = \frac{1}{n\pi} \sin(nt)_{-\pi}^0$$

$$\square b_n = \frac{2}{T} \int_0^T f(t) \cdot \sin(n\omega_0 t) dt$$

$$b_n = \frac{2}{2\pi} \left[\int_{-1}^0 1 \cdot \sin(nt) dT + 0 \right]$$

$$\square \int \sin(u) du = -\cos(u)$$

$$u = nt \quad du = n dt$$

$$\frac{1}{n} \int \sin(nt)(n dt) = -\frac{1}{n} \cos(nt)$$

$$b_n = -\frac{1}{n\pi} \cos(nt)_{-\pi}^0 = \frac{1}{n\pi} [\cos(nt) - 1]$$

$$0, n \text{ es par} \quad \frac{-2}{n\pi}, n \text{ es impar}$$

$$\square f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(t) = \frac{1}{2} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\sin((2n-1)t)}{(2n-1)}$$

$$f(t) = t^2 \quad -\pi < t < \pi$$

$$\square \frac{1}{2} a_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$\frac{1}{2} a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} t^2 dt = \frac{1}{2\pi} \left[\frac{t^3}{3} \right]_{-\pi}^{\pi}$$

$$\frac{1}{2} a_0 = \frac{1}{3(2\pi)} [\pi^3 - (-\pi)^3] = \frac{1}{3(2\pi)} [2\pi^3] = \frac{1}{3} \pi^2$$

$$\square a_n = \frac{2}{T} \int_0^T f(t) \cdot \cos(n\omega_0 t) dt$$

Integración por partes

$$\square \int u dv = uv - \int v du$$

$$u = t^2 \quad dv = \cos(nt) dt$$

$$du = 2t dt \quad v = \frac{1}{n} \sin(nt)$$

$$a_n = \frac{2}{2\pi} \int_{-\pi}^{\pi} t^2 \cos(n\omega_0 t) dt = \frac{1}{\pi} \left[t^2 \frac{1}{n} \sin(nt) - 2 \frac{1}{n} \int_{-\pi}^{\pi} t \sin(nt) dt \right]$$

$$a_n = \frac{1}{n\pi} \left[t^2 \sin(nt) - 2 \int_{-\pi}^{\pi} t \sin(nt) dt \right]$$

$$a_n = \frac{1}{n\pi} \left[t^2 \sin(nt) - 2 \left[-t \cos(nt) - \left(-\frac{1}{n} \int_{-\pi}^{\pi} \cos(nt) dt \right) \right] \right]$$

$$a_n = \frac{1}{n\pi} \left[t^2 \sin(nt) - 2 \frac{1}{n} \left[-t \cos(nt) + \left(\int_{-\pi}^{\pi} \cos(nt) dt \right) \right] \right]$$

$$a_n = \frac{1}{n\pi} \left[t^2 \sin(nt) + \frac{2}{n} t \cos(nt) - \frac{2}{n^2} \sin(nt) \right]_{-\pi}^{\pi}$$

$$a_n = \frac{2}{n^2 \pi} [\pi \cos(n\pi) - (-\pi) \cos(n\pi)] = \frac{2}{n^2 \pi} [2\pi \cos(n\pi)] = \frac{4}{n^2} (-1)^n$$

$$\square b_n = \frac{2}{T} \int_0^T f(t) \cdot \sin(n\omega_0 t) dt$$

$$b_n = \frac{2}{2\pi} \int_{-\pi}^{\pi} t^2 \sin(n\omega_0 t) dt = 0$$

$$\square f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{1}{3} \pi^2 + 4 \sum_{n=1}^{\infty} \frac{(-1)^n \cdot \cos(n\pi)}{n^2}$$

$$f(t) = e^t \quad -\pi < t < \pi$$

$$\frac{1}{2}a_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$\frac{1}{2}a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^t dt = \frac{1}{2\pi} \left[e^t \right]_{-\pi}^{\pi} = \frac{1}{2\pi} [e^{\pi} - e^{-\pi}]$$

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\frac{1}{2}a_0 = \frac{\sinh(\pi)}{\pi}$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cdot \cos(n\omega_0 t) dt$$

$$a_n = \frac{2}{2\pi} \int_{-\pi}^{\pi} e^t \cdot \cos(nt) dt$$

Integración por partes

$$\int u dv = uv - \int u dv$$

$$u = e^t \quad dv = \cos(nt) dt$$

$$du = e^t dt \quad v = \frac{1}{n} \sin(nt)$$

$$a_n = \frac{1}{\pi} \left[e^t \frac{1}{n} \sin(nt) - \frac{1}{n} \int_{-\pi}^{\pi} \sin(nt) e^t dt \right]$$

$$a_n = \frac{1}{n\pi} e^t \sin(nt) - \frac{1}{n\pi} \int_{-\pi}^{\pi} e^t \sin(nt) dt$$

Integración por partes

$$\int u dv = uv - \int u dv$$

$$u = e^t \quad dv = \sin(nt) dt$$

$$du = e^t dt \quad v = -\frac{1}{n} \cos(nt)$$

$$a_n = \frac{1}{n\pi} e^t \sin(nt) - \frac{1}{n\pi} \left(e^t \frac{-1}{n} \cos(nt) - \int_{-\pi}^{\pi} e^t \frac{-1}{n} \cos(nt) dt \right)$$

$$a_n = \frac{1}{n\pi} e^t \sin(nt) + \frac{1}{n^2\pi} e^t \cos(nt) - \frac{1}{n^2\pi} \int_{-\pi}^{\pi} e^t \cos(nt) dt = \frac{1}{n\pi} e^t \sin(nt) + \frac{1}{n^2\pi} e^t \cos(nt) - \frac{1}{n^2\pi} a_n$$

$$a_n + \frac{1}{n^2} a_n = \frac{1}{n\pi} e^t \sin(nt) + \frac{1}{n^2\pi} e^t \cos(nt)$$

$$a_n \left(1 + \frac{1}{n^2} \right) = \frac{1}{n\pi} \left(e^t \sin(nt) + \frac{1}{n} e^t \cos(nt) \right)$$

$$a_n = \frac{n^2}{n^2 + 1} \cdot \frac{1}{n\pi} \left(e^t \sin(nt) + \frac{1}{n} e^t \cos(nt) \right) \Big|_{-\pi}^{\pi} = \frac{n}{(n^2 + 1)\pi} (e^{\pi} (-1)^n - e^{-\pi} (-1)^n)$$

$$a_n = \frac{(-1)^n}{(n^2 + 1)\pi} (e^{\pi} - e^{-\pi}) = \frac{(-1)^n}{(n^2 + 1)\pi} 2\sinh(\pi)$$

$$b_n = \frac{2}{2\pi} \int_{-\pi}^{\pi} e^t \cdot \text{sen}(nt) dt$$

Integración por partes

$$\square \int u dv = uv - \int u dv$$

$$u = e^t \quad dv = \text{sen}(nt) dt$$

$$du = e^t dt \quad v = -\frac{1}{n} \cos(nt)$$

$$b_n = \frac{1}{\pi} \left[e^t \frac{-1}{n} \cos(nt) - \int_{-\pi}^{\pi} \frac{-1}{n} \cos(nt) e^t dt \right] = -\frac{1}{n\pi} e^t \cos(nt) + \frac{1}{n\pi} \int_{-\pi}^{\pi} e^t \cos(nt) dt$$

Integración por partes

$$\square \int u dv = uv - \int u dv$$

$$u = e^t \quad dv = \cos(nt) dt$$

$$du = e^t dt \quad v = \frac{1}{n} \text{sen}(nt)$$

$$b_n = -\frac{1}{n\pi} e^t \cos(nt) + \frac{1}{n\pi} \left(e^t \frac{1}{n} \text{sen}(nt) - \int_{-\pi}^{\pi} e^t \frac{1}{n} \text{sen}(nt) dt \right)$$

$$b_n = -\frac{1}{n\pi} e^t \cos(nt) + \frac{1}{n^2\pi} e^t \text{sen}(nt) - \frac{1}{n^2} \int_{-\pi}^{\pi} b_n$$

$$b_n + \frac{1}{n^2} b_n = -\frac{1}{n\pi} e^t \cos(nt) + \frac{1}{n^2\pi} e^t \text{sen}(nt)$$

$$b_n \left(1 + \frac{1}{n^2} \right) = \frac{1}{n\pi} \left(-e^t \cos(nt) + \frac{1}{n} e^t \text{sen}(nt) \right)$$

$$b_n = \frac{n^2}{n^2 + 1} \cdot \frac{1}{n\pi} \left(-e^t \cos(nt) + \frac{1}{n} e^t \text{sen}(nt) \right) \Big|_{-\pi}^{\pi} = \frac{-n}{(n^2 + 1)\pi} \left(e^{\pi} (-1)^n - e^{-\pi} (-1)^n \right)$$

$$b_n = \frac{-n(-1)^n}{(n^2 + 1)\pi} (e^{\pi} - e^{-\pi}) = \frac{-n(-1)^n}{(n^2 + 1)\pi} 2\text{senh}(\pi)$$

$$\square f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \text{sen}\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{1}{\pi} \text{senh}(\pi) + \sum_{n=1}^{\infty} 2\text{senh}(\pi) \frac{(-1)^n}{(n^2 + 1)\pi} \cdot \cos(n\pi) + \sum_{n=1}^{\infty} 2\text{senh}(\pi) \frac{-n(-1)^n}{(n^2 + 1)\pi} \cdot \text{sen}(n\pi)$$

$$f(t) = A|\text{sen}(\omega_0 t)| \quad 0 < t < \pi$$

$$\square \frac{1}{2}a_0 = \frac{1}{T} \int_0^T f(t) \cdot dt$$

$$\frac{1}{2}a_0 = \frac{1}{2\pi} \int_0^\pi 2A \text{sen}(t) dt = \frac{A}{\pi} [-\cos(t)]_0^\pi$$

$$\frac{1}{2}a_0 = -\frac{A}{\pi} (-1 - (-1)) = \frac{2A}{\pi}$$

$$\square a_n = \frac{2}{T} \int_0^T f(t) \cdot \cos(n\omega_0 t) dt$$

$$a_n = \frac{2}{2\pi} \int_{-\pi}^\pi |2A \text{sen}(t)| \cdot \cos(nt) dt$$

Integración por partes

$$\square \int u dv = uv - \int u dv$$

$$u = \text{sen}(t) \quad dv = \cos(t) dt$$

$$du = \cos(t) dt \quad v = \frac{1}{n} \text{sen}(nt)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^\pi 2A \text{sen}(t) \cos(t) dt = \frac{2A}{\pi} \left[\text{sen}(t) \frac{1}{n} \text{sen}(nt) \Big|_0^\pi - \int_0^\pi \frac{1}{n} \text{sen}(nt) \cos(nt) \right]$$

$$a_n = \frac{-2A}{n\pi} [\text{sen}(nt) \cos(t) dt]$$

Integración por partes

$$\square \int u dv = uv - \int u dv$$

$$u = \cos(t) \quad dv = \text{sen}(nt) dt$$

$$du = -\text{sen}(t) dt \quad v = \frac{-1}{n} \cos(nt)$$

$$a_n = \frac{-2A}{n\pi} \left[\cos(t) \frac{-1}{n} \cos(nt) - \int_0^\pi \frac{1}{n} \cos(nt) \text{sen}(t) dt \right] = \frac{2A}{n\pi} \left[\cos(t) \cos(nt) + \frac{1}{n^2} \cdot \frac{2A}{\pi} \int_0^\pi \cos(nt) \text{sen}(t) dt \right]$$

$$a_n = \frac{2A}{n\pi} \cos(t) \cos(nt) + \frac{1}{n^2} a_n$$

$$a_n \left(1 - \frac{1}{n^2} \right) = \frac{2A}{n\pi} \cos(t) \cos(nt)$$

$$a_n = \frac{n^2}{n^2 - 1} \frac{2A}{n\pi} \cos(t) \cos(nt) \Big|_0^\pi = \frac{2A}{\pi(n^2 - 1)} [(-1) \cos(n\pi) - (1)(1)]$$

$$a_n = \frac{2A}{\pi(n^2 - 1)} [-(-1)^n - 1] = \frac{-2A}{\pi(n^2 - 1)} [(-1)^n + 1] = \frac{2A}{\pi(1 - n^2)} [(-1)^n + 1]$$

$$0, n \text{ es impar} \quad \frac{4A}{\pi(1 - n^2)}, n \text{ es par}$$

$$\square b_n = \frac{2}{T} \int_0^T f(t) \cdot \text{sen}(n\omega_0 t) dt \rightarrow b_n = \frac{2}{2\pi} \int_{-\pi}^\pi |2A \text{sen}(t)| \cdot \text{sen}(nt) dt = 0$$

$$\square f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \text{sen}\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2A}{\pi} + \sum_{n=1}^{\infty} \frac{4A}{\pi(1 - n^2)} \cdot \cos(n\omega_0 t)$$

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

$$a_0 = \frac{1}{2\pi} \left(\int_0^\pi 0 dx + \int_0^\pi x dx \right) = \frac{1}{2\pi} \int_0^\pi x dx$$

$$a_0 = \frac{1}{2\pi} \left(\frac{x^2}{2} \right)_0^\pi = \frac{\pi}{4}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cdot \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_n = \frac{1}{\pi} \left(\frac{x \cos(nx)}{n} + \frac{\cos(nx)}{n} \right)_0^\pi$$

$$a_n = \frac{1}{\pi} \left(0 - 0 + \frac{(-1)^n}{n^2} - \frac{1}{n^2} \right)$$

$$a_n = \frac{(-1)^n - 1}{n^2 \pi}$$

$$0, n \text{ par} \quad -\frac{2}{n^2 \pi}, \text{ impar}$$

$$a_{2n} = 0 \quad \forall n$$

$$a_{2n-1} = -\frac{2}{(2n-1)^2 \pi} \quad \forall n$$

$$b_n = \int_{-1}^0 -t \cdot \sin(n\pi t) dt$$

$$b_n = \frac{1}{\pi} \left(\int_0^\pi x \sin(nx) dx \right)$$

$$b_n = \frac{1}{\pi} \left[-\frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^\pi = -\frac{x \cos(n\pi)}{n}$$

Luego coeficiente es :

$$b_n = \frac{(-1)^{n+1}}{n}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{\pi}{8} \sum_{n=1}^{\infty} \left[-\frac{2}{(2n-1)^2 \pi} \cos((2n-1)x) + \frac{(-1)^{n+1}}{n} \sin(nx) \right]$$

EJERCICIOS DE SERIES DE POTENCIA

FORMULAS

$$y = \sum_{n=0}^{\infty} C_n x^n$$

$$y' = \sum_{n=1}^{\infty} n C_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2}$$

1. $y' = 2xy$

$$y' - 2xy = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - 2x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - \sum_{n=0}^{\infty} 2 C_n x^{n+1} = 0$$

$$0 C_0 x^0 + C_1 x^{n-1} + \sum_{n=2}^{\infty} n C_n x^{n-1} - \sum_{n=0}^{\infty} 2 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} (n+2) C_{n+2} x^{n+1} - \sum_{n=0}^{\infty} 2 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} [(n+2) C_{n+2} - 2 C_n] x^{n+1} = 0$$

$$(n+2) C_{n+2} - 2 C_n = 0 \quad C_1 = 0$$

$$C_{n+2} = \frac{2}{(n+2)} C_n$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = 2 C_0 / 2 = C_0$$

$$n = 1 \quad C_3 = 2 C_1 / 3 = 0$$

$$n = 2 \quad C_4 = 2 C_2 / 4 = 1/2 C_0$$

Si n es impar, $C_n = 0$

$$n = 2k + 1 \quad C_n = 0 \quad C_{2k+1} = 0$$

Si n es par, $C_n = 0$

$$n = 2k \quad C_n = \frac{1}{2 * 3 * \dots * k} C_0 \quad C_{2k} = \frac{1}{2 * 3 * 4 * \dots * k} C_0 \quad C_{2k} = \frac{1}{k!} C_0$$

$$y = C_0 x^{2*0} + C_0 x^{2*1} + C_0 \frac{1}{2} x^{2*2} + C_0 \frac{1}{3} x^{2*3} + \dots \dots \dots$$

$$y = \sum_{n=0}^{\infty} \frac{1}{k!} C_0 x^{2k} = C_0 \sum_{n=0}^{\infty} \frac{(x^2)^k}{k!}$$

FORMULA

$$\sum_{n=0}^{\infty} (n+K) f(n) = \sum_{n=0}^{\infty} (n+K) f(n+K)$$

FORMULA

$$y = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \dots \dots \dots$$

FORMULA

$$e^v = \sum_{n=0}^{\infty} \frac{v^n}{n!}$$

$$y = C_0 * e^{x^2} \quad y = C * e^{x^2}$$

2. $xy' = (x+1)y$

$$\begin{aligned}
 xy' - xy - y &= 0 \\
 x \sum_{n=1}^{\infty} n C_n x^{n-1} - x \sum_{n=0}^{\infty} C_n x^n - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 \sum_{n=1}^{\infty} n C_n x^n - \sum_{n=0}^{\infty} C_n x^{n+1} - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 \sum_{n=0}^{\infty} (n+1) C_{n+1} x^{n+1} - \sum_{n=0}^{\infty} C_n x^{n+1} - C_0 - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 -C_0 + \sum_{n=0}^{\infty} [(n+1) C_{n+1} - C_n - C_{n+1}] x^{n+1} &= 0 \\
 (n+2) C_{n+2} - 2 C_n &= 0 \qquad -C_0 = 0 \quad C_0 = 0 \\
 C_{n+1} &= \frac{C_n}{n}
 \end{aligned}$$

Remplacemos n en la ecuación obtenida

$n = 0 \quad C_1 = C_0/0 = 0$

$n = 1 \quad C_2 = C_1/1 = 0 \quad C_2 = C_1$

$n = 2 \quad C_3 = C_2/2 = 1/2 C_1$

$C_n = \frac{1}{2 * 3 * 4 * 5 * \dots * (n-1)} C_1 \quad C_n = \frac{1}{(n-1)!} C_1$

$y = \sum_{n=0}^{\infty} \frac{1}{(n)!} C_1 x^{n+1} = C_1 \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n)!}$

$$C_1 \sum_{n=0}^{\infty} \frac{x^n}{(n)!} = C * e^{x^2}$$

FORMULA

$$y = \sum_{n=1}^{\infty} \frac{1}{(n-1)!} C_1 x^n$$

$$y = \sum_{n=1}^{\infty} C_n x^n$$

3. $(2-x)y' = y$

$$\begin{aligned}
 2y' - xy - y &= 0 \\
 2 \sum_{n=1}^{\infty} n C_n x^{n-1} - x \sum_{n=0}^{\infty} C_n x^n - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 \sum_{n=1}^{\infty} 2 C_n x^n - \sum_{n=0}^{\infty} C_n x^{n+1} - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 \sum_{n=0}^{\infty} 2(n+1) C_{n+1} x^{n+1} - \sum_{n=0}^{\infty} C_n x^{n+1} - C_0 - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 -C_0 + \sum_{n=0}^{\infty} [2(n+1) C_{n+1} - n C_n - C_n] x^{n+1} &= 0 \\
 2(n+1) C_{n+1} - n C_n - C_n &= 0 \\
 C_{n+1} &= \frac{1}{2} C_n
 \end{aligned}$$

Remplacemos n en la ecuación obtenida

$n = 0 \quad C_1 = 1/2 C_0$

$n = 1 \quad C_2 = 1/2^2 C_0 \quad C_2 = C_1$

$$n = 2 \quad C_3 = 1/3^2 C_0$$

$$C_n = \frac{1}{2^n} C_0$$

$$y = \sum_{n=0}^{\infty} \frac{1}{2^n} C_0 x^n = C_0 \sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n$$

$$y = C_0 \frac{1}{1 - \frac{x}{2}} \quad \left| \frac{x}{2} \right| < 1$$

$$y = C_0 \frac{1}{\frac{2-x}{2}} = \frac{2C_0}{2-x}$$

$$y = \frac{C}{2-x}$$

FORMULA

$$\sum_{n=1}^{\infty} a^n = \frac{1}{1-a} \quad \text{si } |a| < 1$$

$$|x| < a \quad -a < x < a$$

$$4. \quad y' + 4xy = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 4x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} 4 C_n x^{n+1} = 0$$

$$0 C_0 x^0 + C_1 x^{n-1} + \sum_{n=2}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} 4 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} (n+2) C_{n+2} x^{n+1} + \sum_{n=0}^{\infty} 4 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} [(n+2) C_{n+2} - 4 C_n] x^{n+1} = 0$$

$$(n+2) C_{n+2} - 4 C_n = 0 \quad C_1 = 0$$

$$C_{n+2} = \frac{4}{(n+2)} C_n$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = 2 C_0$$

$$n = 1 \quad C_3 = 4/3 C_1 = 0$$

$$n = 2 \quad C_4 = 2 C_0$$

$$n = 1 \quad C_3 = 8/9 C_1 = 0$$

$$y = C_0 (2x^2 + 2x^4 + 2x^6 + \dots \dots \dots)$$

$$5. \quad (x-3)y' + 2y = 0$$

$$xy' - 3y' + 2y = 0$$

$$x \sum_{n=0}^{\infty} n C_n x^{n-1} - 3 \sum_{n=1}^{\infty} n C_n x^{n-1} + 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^n - \sum_{n=0}^{\infty} 3n C_n x^{n-1} + \sum_{n=0}^{\infty} 2 C_n x^n = 0$$

$$\sum_{n=0}^{\infty} n C_n x^n - \sum_{n=0}^{\infty} 3(n+1) C_{(n+1)} x^n + \sum_{n=0}^{\infty} 2 C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [nC_n - 3(n+1)C_{n+1} + 2C_n]x^n = 0$$

$$nC_n - 3(n+1)C_{n+1} + 2C_n = 0$$

$$C_{n+1} = \frac{(n+2)}{3(n+1)} C_n$$

Remplacemos n en la ecuación obtenida

$$n=0 \quad C_1 = \frac{2}{3} C_0$$

$$n=1 \quad C_2 = \frac{1}{3} C_0$$

$$n=2 \quad C_3 = \frac{4}{27} C_0$$

$$C_n = \frac{n+1}{3^n} C_0$$

$$y = \sum_{n=0}^{\infty} \frac{n+1}{3^n} C_0 x^n = C_0 \sum_{n=0}^{\infty} (n+1) \left(\frac{x}{3}\right)^n$$

$$y = C_0 \frac{1}{\left(\frac{x}{3} - 1\right)^2} \quad \left|\frac{x}{3}\right| < 1$$

$$y = \frac{9C_0}{(x-3)^2}$$

FORMULA

$$\sum_{n=1}^{\infty} (n+1)x^n = \frac{1}{(x-1)^2} \quad \text{si } |1| = 1$$

$$6. \quad xy' + y = 0$$

$$x \sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} nC_n x^n + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [nC_n + C_n] x^n = 0$$

$$nC_n + C_n = 0$$

$$C_n = 0$$

$$y = \sum_{n=0}^{\infty} 0x^n = 0 \quad y = 0$$

$$7. \quad x^2 y' = y - x - 1$$

$$x^2 y' - y + x + 1 = 0$$

$$\begin{aligned}
x^2 \sum_{n=1}^{\infty} n C_n x^{n-1} - \sum_{n=0}^{\infty} C_n x^n + x + 1 &= 0 \\
x + 1 - \sum_{n=0}^{\infty} n C_n x^{n+1} - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
x + 1 - \sum_{n=0}^{\infty} n C_n x^{n+1} - (C_0 + \sum_{n=1}^{\infty} C_n x^n) &= 0 \\
x + 1 - \sum_{n=0}^{\infty} n C_n x^{n+1} - C_0 - \sum_{n=1}^{\infty} C_{n+1} x^{n+1} &= 0 \\
1 - C_0 + x + \sum_{n=0}^{\infty} [-C_{n+1} + n C_n] x^{n+1} &= 0 \\
1 - C_0 + (1 - C_1)x + \sum_{n=0}^{\infty} [-C_{n+1} + n C_n] x^{n+1} &= 0 \\
1 - C_0 = 0 & \quad C_0 = 1 \\
1 - C_1 = 0 & \quad C_1 = 1 \\
C_{n+1} = n C_n &
\end{aligned}$$

Remplacemos n en la ecuación obtenida

$$n = 1 \quad C_2 = C_1$$

$$n = 2 \quad C_3 = 2 C_2$$

$$n = 3 \quad C_4 = 3 C_3$$

$$y = 1 + x + \sum_{n=2}^{\infty} (n-1)! x^n$$

$$8. \quad y'' + y = 0$$

$$\begin{aligned}
\sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} + \sum_{n=0}^{\infty} C_n x^n &= 0 \\
\sum_{n=2}^{\infty} (n+2)(n-1) C_{n+2} x^n + \sum_{n=0}^{\infty} C_n x^n &= 0 \\
\sum_{n=0}^{\infty} [(n+2)(n+1) C_{n+2} + C_n] x^n &= 0 \\
(n+2)(n+1) C_{n+2} + C_n &= 0 \\
C_{n+2} &= -\frac{1}{(n+2)(n+1)} C_n
\end{aligned}$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = -1/2 C_0$$

$$n = 1 \quad C_3 = -1/3 * 2 C_1$$

$$n = 2 \quad C_4 = 1/4 * 3 * 2 C_0$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} + C_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$y = C_0 \cos x + C_1 \sin x$$

$$9. \quad y'' = 4y$$

$$y'' - 4y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - 4 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} (n+2)(n-1)C_{n+2} x^n - \sum_{n=0}^{\infty} 4C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} - 4C_n] x^n = 0$$

$$(n+2)(n+1)C_{n+2} + C_n = 0$$

$$C_{n+2} = \frac{4}{(n+2)(n+1)} C_n$$

Reemplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = 4/2 C_0$$

$$n = 1 \quad C_3 = 4/3 * 2 C_1$$

$$n = 2 \quad C_4 = 4^2/4 * 3 * 2 C_0$$

$$n = 3 \quad C_5 = 4^2/5 * 4 * 3 * 2 C_1$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{4^n x^{2n}}{(2n)!} + C_1 \sum_{n=0}^{\infty} \frac{4^n x^{2n+1}}{(2n+1)!}$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{2^{2n} x^{2n}}{(2n)!} + C_1 \sum_{n=0}^{\infty} \frac{2^{2n} x^{2n+1}}{(2n+1)!}$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{2x^{2n}}{(2n)!} + C_1/2 \sum_{n=0}^{\infty} \frac{2x^{2n+1}}{(2n+1)!}$$

$$y = C_0 \cosh(2x) + \frac{C_1}{2} \sinh(2x)$$

FORMULA

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

FORMULA

$$\cosh(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$\sinh(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

10. $y'' + y = x$

$$y'' + y - x = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=0}^{\infty} C_n x^n - x = 0$$

$$-x + \sum_{n=2}^{\infty} (n+2)(n-1)C_{n+2} x^n + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$-x + \sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} - C_n] x^n = 0$$

$$-x + 2C_2 + C_0 + (6C_3 + C_1)x + \sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} - C_n] x^n = 0$$

$$2C_2 + C_0 = 0$$

$$C_2 = -1/2 C_0$$

$$6C_3 + C_1 - 1 = 0$$

$$C_3 = \frac{1}{3!} C_1$$

$$(n+2)(n+1)C_{n+2} + C_n = 0$$

$$C_{n+2} = \frac{1}{(n+2)(n+1)} C_n$$

Remplacemos n en la ecuación obtenida

$$n = 2 \quad C_4 = \frac{1}{4!} C_0$$

$$n = 3 \quad C_5 = \frac{1}{5!} C_1$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} + C_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$y = C_0 \cos x + C_1 \sin x + x$$

11. $y'' + 4y = 0 \quad y(0) = 0 \quad y'(0) = 3$

$$y'' + 4y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + 4 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} (n+2)(n-1)C_{n+2} x^n + \sum_{n=0}^{\infty} 4C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} + 4C_n] x^n = 0$$

$$(n+2)(n+1)C_{n+2} + 4C_n = 0$$

$$y(0) = 0 \quad y'(0) = 3$$

$$C_0 = 0 \quad C_1 = 3$$

$$C_{n+2} = -\frac{4}{(n+2)(n+1)} C_n$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = 0$$

$$n = 1 \quad C_3 = 4 * 3 / 3! C_1$$

$$n = 2 \quad C_4 = 0$$

$$n = 3 \quad C_4 = 4^2 * 3 / 5! C_1$$

$$y = \sum_{n=0}^{\infty} \frac{(-1)^n * 4^n * 3}{(2n+1)!} x^{2n+1}$$

$$y = \frac{3}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (2x)^{2n+1}$$

$$y = \frac{3}{2} \text{sen}(2x)$$

$$12. \quad y'' + y' - 2y = 0 \quad y(0) = 1 \quad y'(0) = 2$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} nC_n x^{n-1} - 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n-1)C_{n+2} x^n + \sum_{n=0}^{\infty} (n-1)C_{n+1} x^n - \sum_{n=0}^{\infty} 2C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} + (n+1)C_{n+1} - 2C_n] x^n = 0$$

$$y(0) = 1 \quad y'(0) = 2$$

$$C_0 = 1 \quad C_1 = 2$$

$$(n+2)(n+1)C_{n+2} + (n+1)C_{n+1} - 2C_n = 0$$

$$C_{n+2} = -\frac{2C_n - (n+1)C_{n+1}}{(n+2)(n+1)}$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = 2$$

$$n = 1 \quad C_3 = -4/3$$

$$n = 2 \quad C_4 = 2/3$$

$$y = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \dots \dots \dots$$

$$y = 1 - 2x + 2x^2 - C_0 \frac{4}{3}x^3 + C_0 \frac{2}{3}x^4 + \dots \dots \dots$$

13. $y'' = xy$

$$y'' - yx = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$2C_2 + \sum_{n=3}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$2C_2 + \sum_{n=0}^{\infty} (n+3)(n+2)C_{n+3} x^{n+1} + \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$2C_2 + \sum_{n=0}^{\infty} [(n+3)(n+2)C_{n+3} - C_n] x^n = 0$$

$$2C_2 = 0 \quad C_2 = 0$$

$$(n+3)(n+1)C_{n+3} + C_n = 0$$

$$C_{n+3} = -\frac{1}{(n+2)(n+3)} C_n$$

Remplacemos n en la ecuación obtenida t

$$n = 0 \quad C_3 = \frac{4}{3!} C_0$$

$$n = 1 \quad C_4 = \frac{2}{4!} C_1$$

$$n = 2 \quad C_5 = 0$$

$$n = 3 \quad C_6 = \frac{4}{6!} C_0$$

$$n = 4 \quad C_7 = \frac{2 \cdot 5}{7!} C_1$$

$$n = 5 \quad C_8 = 0$$

$$C_{3n+2} = 0 \quad C_{3n} = \frac{1 \cdot 4 \cdot 7 \dots (3n-2)}{(3n)!} C_0$$

$$C_{3n+1} = -\frac{2 \cdot 5 \dots (3n-1)}{(3n+1)!} C_1$$

$$y = C_0 \left(1 + \frac{1}{3!}x^3 + \frac{4}{6!}x^6 + \dots \right) + C_1 \left(x + \frac{1}{4!}x^4 + \dots \right)$$

14. $y'' - 2xy' + 4y = 0$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - 2x \sum_{n=1}^{\infty} nC_n x^{n-1} + 4 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n-1)C_{n+2}x^n + \sum_{n=0}^{\infty} 2nC_nx^n - \sum_{n=0}^{\infty} 4C_nx^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} + 2nC_n - 4C_n]x^n = 0$$

$$(n+2)(n+1)C_{n+2} + 2nC_n - 4C_n = 0$$

$$C_{n+2} = -\frac{2(n-2)C_n}{(n+2)(n+1)}$$

Remplacemos n en la ecuación obtenida t

$$n=0 \quad C_2 = -2C_0$$

$$n=1 \quad C_3 = -\frac{1}{3}C_1$$

$$n=2 \quad C_4 = 0$$

$$n=3 \quad C_5 = -\frac{1}{30}C_0$$

$$y = C_0(1 - 2x^2) + C_1\left(x + \frac{1}{3}x^3 + \frac{1}{30}x^5 + \frac{1}{210}x^7 \dots\right)$$

$$15. \quad xy'' + y' + xy = 0$$

$$x \sum_{n=2}^{\infty} n(n-1)C_nx^{n-2} + \sum_{n=1}^{\infty} nC_nx^{n-1} + x \sum_{n=0}^{\infty} C_nx^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_nx^{n-1} + \sum_{n=1}^{\infty} nC_nx^{n-1} + \sum_{n=0}^{\infty} C_nx^{n+1} = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_nx^{n-1} + C_1 + \sum_{n=2}^{\infty} nC_nx^{n-1} + \sum_{n=0}^{\infty} C_nx^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} (n+2)(n-1)C_{n+2}x^{n+1} + \sum_{n=0}^{\infty} (n-1)C_{n+2}x^{n+1} + \sum_{n=0}^{\infty} C_nx^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} + (n-1)C_{n+2} + C_n]x^{n+1} = 0$$

$$C_1 = 0$$

$$(n+2)(n+1)C_{n+2} + (n-1)C_{n+2} + C_n = 0$$

$$C_{n+2} = -\frac{C_n}{(n+2)^2}$$

$$n=0 \quad C_2 = -\frac{1}{2^2}C_0$$

$$n=1 \quad C_3 = 0$$

$$n=2 \quad C_4 = -\frac{1}{2^2} \cdot \frac{1}{4^2} C_0$$

$$n = 3 \quad C_5 = 0$$

$$C_{2n} = -\frac{(-1)^n C_0}{2^{2n} (n!)^2}$$

$$y = C_0 \sum_{n=2}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n} (n!)^2}$$

$$16. x^2 y'' + x^2 y' + y = 0$$

$$x^2 \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} + x^2 \sum_{n=1}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) C_n x^n + \sum_{n=1}^{\infty} n C_n x^{n+1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n(n-1) C_n x^n + \sum_{n=0}^{\infty} n C_n x^{n+1} + C_0 + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$C_0 + \sum_{n=0}^{\infty} (n+1) n C_{n+1} x^{n+1} + \sum_{n=0}^{\infty} n C_n x^{n+1} + \sum_{n=0}^{\infty} C_{n+1} x^{n+1} = 0$$

$$C_0 + \sum_{n=0}^{\infty} [(n+1) n C_{n+1} + n C_n + C_{n+1}] x^{n+1} = 0$$

$$(n+1) n C_{n+1} + n C_n + C_{n+1} = 0$$

$$C_{n+1} = -\frac{n C_n}{n^2 + n + 1}$$

$$n = 0 \quad C_1 = 0$$

$$n = 1 \quad C_2 = 0$$

$$y = 0$$

$$17. 3xy'' + (2-x)y' - 2y = 0$$

$$3x \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} + 2 \sum_{n=1}^{\infty} n C_n x^{n-1} - x \sum_{n=1}^{\infty} n C_n x^{n-1} + 2 \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} 3n(n-1) C_n x^{n+1} + \sum_{n=1}^{\infty} 2n C_n x^{n+1} - \sum_{n=1}^{\infty} n C_n x^n + \sum_{n=1}^{\infty} 2C_n x^n = 0$$

$$\sum_{n=0}^{\infty} 3n(n-1) C_n x^n + \sum_{n=0}^{\infty} 2(n+1) C_{n+1} x^n - \sum_{n=0}^{\infty} n C_n x^n + \sum_{n=0}^{\infty} 2C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [3n(n-1) C_n + 2(n+1) C_{n+1} + n C_n + 2C_n] x^n = 0$$

$$3n(n-1)C_n + 2(n+1)nC_{n+1} + nC_n + 2C_n = 0$$

$$C_{n+1} = -\frac{(n+2)C_n}{(n+1)(3n+2)}$$

$$n=0 \quad C_2 = C_0$$

$$n=1 \quad C_3 = \frac{3}{10}C_0$$

$$n=2 \quad C_4 = \frac{1}{20}C_0$$

$$y = C_0 \left(1 + x + \frac{3}{10}x^2 + \frac{1}{20}x^3 + \frac{1}{176}x^4 + \dots \right)$$

$$18. y' = y$$

$$y' - y = 0$$

$$\sum_{n=1}^{\infty} nC_n x^{n-1} - \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} (n+1)C_{n+1}x^n - \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+1)C_{n+1} - C_n]x^n = 0$$

$$(n+1)C_{n+1} - C_n = 0$$

$$C_{n+1} = \frac{C_n}{(n+1)}$$

$$n=0 \quad C_1 = \frac{1}{1}C_0$$

$$n=1 \quad C_2 = \frac{1}{2}C_0$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$y = C_0 e^x$$

$$19. y'' - 2xy' + y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - 2x \sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - \sum_{n=1}^{\infty} 2nC_n x^n + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n+1)C_{n+2}x^n - \sum_{n=0}^{\infty} 2nC_n x^n + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} - 2nC_n + C_n]x^n = 0$$

$$C_0 = 0$$

$$(n+2)(n+1)C_{n+2} - 2nC_n + C_n = 0$$

$$C_{n+2} = -\frac{(2n-1)}{(n+2)(n+1)}C_n$$

$$n=0 \quad C_2 = -1/2$$

$$n=1 \quad C_3 = 0$$

$$n=2 \quad C_4 = -1/8$$

$$n=4 \quad C_6 = -7/240$$

$$y = C_0 \left(1 - \frac{1}{2}x^2 - \frac{1}{8}x^4 + \dots \right) + C_1 \left(x + \frac{1}{6}x^3 + \dots \right)$$

$$20. (x^2 + 2)y'' + 3xy' - y = 0$$

$$x^2 \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + 3x \sum_{n=1}^{\infty} nC_n x^{n-1} - \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} 2n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} 3nC_n x^n - \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} (n+2)(n+1)C_{n+2} x^n + \sum_{n=1}^{\infty} 3nC_n x^n - \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [n(n-1)C_n + (n+2)(n+1)C_{n+2} + 3nC_n - C_n]x^n = 0$$

$$n(n-1)C_n + (n+2)(n+1)C_{n+2} + 3nC_n - C_n = 0$$

$$C_{n+2} = -\frac{(-3n-1-(n-1))}{(n+2)(n+1)}C_n$$

$$n=0 \quad C_2 = 1/4 C_0$$

$$n=1 \quad C_3 = -1/6 C_1$$

$$n=2 \quad C_4 = -7/96 C_0$$

$$n=4 \quad C_6 = 7/120 C_1$$

$$y = C_0 \left(1 - \frac{1}{4}x^2 - \frac{7}{96}x^4 + \dots \right) + C_1 \left(x + \frac{1}{6}x^3 + \frac{7}{120}x^5 + \dots \right)$$

$$21. (x^2 + 1)y'' - xy' + y = 0$$

$$x^2 \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - x \sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} 2n(n-1)C_n x^{n-2} + \sum_{n=0}^{\infty} nC_n x^n - \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} (n+2)(n+1)C_{n+2} x^n + \sum_{n=1}^{\infty} nC_n x^n - \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [n(n-1)C_n + (n+2)(n+1)C_{n+2} + nC_n - C_n]x^n = 0$$

$$n(n-1)C_n + (n+2)(n+1)C_{n+2} + nC_n - C_n = 0$$

$$C_{n+2} = -\frac{n(n-1)}{(n+2)(n+1)}C_{n+1} - \frac{(n-1)}{(n+2)(n+1)}C_n$$

$$y = 6x - 2\left(1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 \dots\right)$$

$$22. (x^2 + 1)y'' - 6y = 0$$

$$x^2 \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - 6 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} 2n(n-1)C_n x^{n-2} - \sum_{n=0}^{\infty} 6C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} (n+2)(n+1)C_{n+2} x^n - \sum_{n=1}^{\infty} 6C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [n(n-1)C_n + (n+2)(n+1)C_{n+2} - 6C_n]x^n = 0$$

$$n(n-1)C_n + (n+2)(n+1)C_{n+2} - 6C_n = 0$$

$$C_{n+1} = -\frac{n(n-1)}{(n+1+6)}C_n$$

$$n = 0 \quad C_1 = 0$$

$$n = 1 \quad C_2 = 0$$

$$y = 0$$

$$23. y'' + xy' + 8y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + x \sum_{n=1}^{\infty} nC_n x^{n-1} + 8 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} nC_n x^n + \sum_{n=0}^{\infty} 8C_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n+1)C_{n+2} x^n + \sum_{n=0}^{\infty} nC_n x^n + \sum_{n=0}^{\infty} 8C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} - 2nC_n + C_n]x^n = 0$$

$$(n+2)(n+1)C_{n+2} + nC_n + C_n = 0$$

$$C_{n+2} = -\frac{(-n-8)}{(n+2)(n+1)}C_n$$

$$n=0 \quad C_2 = -8/2 C_0$$

$$n=1 \quad C_3 = -9/3! C_1$$

$$n=2 \quad C_4 = 80/4! C_0$$

$$n=4 \quad C_6 = 99/5! C_1$$

$$y = C_0 \left(1 - \frac{8}{2}x^2 - \frac{8}{4!}x^4 + \dots \right) + C_1 \left(x + \frac{9}{3!}x^3 + \frac{9}{5!}x^5 \dots \right)$$

$$24. \quad y'' + 5y = 0 \quad y(0) = 0 \quad y'(0) = 1$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + 5 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} (n+2)(n-1)C_{n+2}x^n + \sum_{n=0}^{\infty} 5C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} + 5C_n]x^n = 0$$

$$(n+2)(n+1)C_{n+2} + 5C_n = 0$$

$$y(0) = 0 \quad y'(0) = 1$$

$$C_0 = 0 \quad C_1 = 1$$

$$C_{n+2} = -\frac{5}{(n+2)(n+1)}C_n$$

Reemplacemos n en la ecuación obtenida

$$n=0 \quad C_2 = 0$$

$$n=1 \quad C_3 = 5/3! C_1$$

$$n=2 \quad C_4 = 0$$

$$n=3 \quad C_5 = 5/5! C_1$$

$$y = C_1 \left(x + \frac{5}{3!}x^3 + \frac{5}{5!}x^5 + \frac{5}{7!}x^7 \dots \right)$$

25. $y' + 3xy = 0$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 3x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} 3 C_n x^{n+1} = 0$$

$$0 C_0 x^0 + C_1 x^{n-1} + \sum_{n=2}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} 3 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} (n+2) C_{n+2} x^{n+1} + \sum_{n=0}^{\infty} 3 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} [(n+2) C_{n+2} - 3 C_n] x^{n+1} = 0$$

$$(n+2) C_{n+2} - 3 = 0 \qquad C_1 = 0$$

$$C_{n+2} = \frac{3}{(n+2)} C_n$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = -\frac{3}{2} C_0$$

$$n = 1 \quad C_3 = 0$$

$$n = 2 \quad C_4 = -\frac{3^2}{8} C_0$$

$$n = 1 \quad C_3 = 0$$

$$y = C_0 \left(-\frac{3}{2} x^2 + -\frac{3^2}{8} x^4 + \dots \dots \dots \right)$$

26. $y' + xy = 0$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$k = n - 1 \quad k = n + 1$$

$$n = k + 1 \quad n = k - 1$$

$$\text{Si } n = 1 \rightarrow k = 0 \quad n = 0 \rightarrow k = 1$$

$$\sum_{k=0}^{\infty} (k+1) C_{k+1} x^k + x \sum_{k=1}^{\infty} C_{k-1} x^k = 0$$

$$C_1 + \sum_{k=1}^{\infty} (k+1) C_{k+1} x^k + x \sum_{k=1}^{\infty} C_{k-1} x^k = 0$$

$$C_1 = 0$$

$$\sum_{k=1}^{\infty} [(k+1) C_{k+1} + C_{k-1}] x^k = 0$$

$$(k+1) C_{k+1} + C_{k-1} = 0$$

$$C_{k+1} = -\frac{C_{k-1}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 1 \quad C_2 = -C_0/2$$

$$k = 2 \quad C_3 = 0$$

$$k = 3 \quad C_4 = -C_0/8$$

$$k = 4 \quad C_5 = 0$$

$$k = 5 \quad C_6 = -C_0/48$$

$$y = C_0 \left(1 - \frac{x^2}{2} + \frac{x^4}{8} - \frac{x^6}{48} + \frac{x^8}{384} + \dots \right)$$

$$y = C_0 \left(1 - \frac{x^2}{2^1 * 1!} + \frac{x^4}{2^2 * 2!} - \frac{x^6}{2^3 * 3!} + \frac{x^8}{2^4 * 4!} + \dots \right)$$

27. $y' - x^2 y = 0$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - x^2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - \sum_{n=0}^{\infty} C_n x^{n+2} = 0$$

$$k = n - 1 \quad k = n + 2$$

$$n = k + 1 \quad n = k - 2$$

$$\text{Si } n = 1 \rightarrow k = 0 \quad n = 0 \rightarrow k = 2$$

$$\sum_{k=0}^{\infty} (k+1) C_{k+1} x^k - \sum_{k=2}^{\infty} C_{k-2} x^k = 0$$

$$C_1 + 2C_2 x + \sum_{k=2}^{\infty} (k+1) C_{k+1} x^k - \sum_{k=2}^{\infty} C_{k-2} x^k = 0$$

$$C_1 = 0$$

$$2C_2 = 0$$

$$C_2 = 0$$

$$\sum_{k=2}^{\infty} [(k+1) C_{k+1} - C_{k-2}] x^k = 0$$

$$(k+1) C_{k+1} - C_{k-2} = 0$$

$$C_{k+1} = -\frac{C_{k-2}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 2 \quad C_3 = C_0/3$$

$$k = 3 \quad C_4 = 0$$

$$k = 4 \quad C_5 = 0$$

$$k = 5 \quad C_6 = C_0/18$$

$$k = 6 \quad C_7 = 0$$

$$k = 7 \quad C_8 = 0$$

$$k = 8 \quad C_9 = C_0/162$$

$$y = C_0 \left(1 + \frac{x^3}{3} + \frac{x^6}{18} + \frac{x^9}{162} + \cdots \right)$$

$$y = C_0 \left(1 + \frac{x^3}{3^1 * 1!} + \frac{x^6}{3^2 * 2!} + \frac{x^9}{3^3 * 3!} + \cdots \right)$$

28. $y' + 3x^3y = 0$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 3x^3 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 3 \sum_{n=0}^{\infty} C_n x^{n+3} = 0$$

$$k = n - 1 \quad k = n + 3$$

$$n = k + 1 \quad n = k - 3$$

$$\text{Si } n = 1 \rightarrow k = 0 \quad n = 0 \rightarrow k = 3$$

$$\sum_{k=0}^{\infty} (k+1) C_{k+1} x^k + 3 \sum_{k=3}^{\infty} C_{k-3} x^k = 0$$

$$C_1 + 2C_2x + 3C_3x^2 + \sum_{k=3}^{\infty} (k+1) C_{k+1} x^k + 3 \sum_{k=3}^{\infty} C_{k-3} x^k = 0$$

$$C_1 = 0$$

$$2C_2x = 0$$

$$C_2 = 0$$

$$3C_3x = 0$$

$$C_3 = 0$$

$$\sum_{k=3}^{\infty} [(k+1) C_{k+1} + 3C_{k-3}] x^k = 0$$

$$(k+1) C_{k+1} + 3C_{k-3} = 0$$

$$C_{k+1} = -\frac{3C_{k-3}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 3 \quad C_4 = -\frac{3C_0}{4}$$

$$k = 4 \quad C_5 = 0$$

$$k = 5 \quad C_6 = 0$$

$$k = 6 \quad C_7 = 0$$

$$k = 7 \quad C_8 = \frac{9C_0}{32}$$

$$k = 8 \quad C_9 = 0$$

$$k = 9 \quad C_{10} = 0$$

$$k = 10 \quad C_{11} = 0$$

$$y = C_0 \left(1 - \frac{3}{4}x^4 + \frac{9}{32}x^8 + \dots \right)$$

$$y = C_0 \left(1 - \frac{3^1 x^4}{2^{2*1} * 1!} + \frac{3^2 x^8}{2^{2*2} * 2!} + \dots \right)$$

29. $y' + 3y = 0$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 3 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\begin{array}{cc} k = n - 1 & k = 0 \\ n = k + 1 & n = 0 \end{array}$$

Si $n = 1 \rightarrow k = 0$

$$\sum_{k=0}^{\infty} (k+1) C_{k+1} x^k + 3 \sum_{k=3}^{\infty} C_k x^k = 0$$

$$\sum_{k=3}^{\infty} [(k+1) C_{k+1} + 3 C_k] x^k = 0$$

$$(k+1) C_{k+1} + 3 C_k = 0$$

$$C_{k+1} = \frac{-3 C_k}{k+1}$$

Remplacemos k en la ecuación obtenida

$k = 0 \quad C_1 = -3 C_0$

$k = 1 \quad C_2 = \frac{9}{2} C_0$

$k = 2 \quad C_3 = -\frac{9}{2} C_0$

$k = 3 \quad C_4 = \frac{27}{8} C_0$

$$y = C_0 \left(1 - 3x + \frac{9}{2} x^2 - \frac{9}{2} x^3 + \frac{27}{8} x^4 + \dots \right)$$

30. $y'' - 2xy = 0$

$$\sum_{n=2}^{\infty} (n-1)n C_n x^{n-2} - 2x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} (n-1)n C_n x^{n-2} - 2 \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$\begin{array}{cc} k = n - 2 & k = n + 1 \\ n = k + 2 & n = k - 1 \end{array}$$

Si $n = 2 \rightarrow k = 0 \quad n = 0 \rightarrow k = 1$

$$\sum_{k=0}^{\infty} (k+1)(k+2) C_{k+2} x^k - 2 \sum_{k=3}^{\infty} C_{k-1} x^k = 0$$

$$2 C_2 + \sum_{k=1}^{\infty} (k+1)(k+2) C_{k+2} x^k - 2 \sum_{k=1}^{\infty} C_{k-1} x^k = 0$$

$$2 C_2 = 0$$

$$C_2 = 0$$

$$\sum_{k=1}^{\infty} [(k+1)(k+2) C_{k+2} - 2 C_{k-1}] x^k = 0$$

$$(k+1)(k+2) C_{k+2} - 2 C_{k-1} = 0$$

$$C_{k+2} = \frac{2C_{k-1}}{(k+1)(k+2)}$$

Remplacemos k en la ecuación obtenida

$$k = 1 \quad C_3 = 2C_0/2 * 3$$

$$k = 2 \quad C_4 = 2C_1/3 * 4$$

$$k = 3 \quad C_5 = 0$$

$$k = 4 \quad C_6 = 2 * 2C_0/2 * 3 * 5 * 6$$

$$k = 5 \quad C_7 = 2 * 2C_1/3 * 4 * 6 * 7$$

$$k = 6 \quad C_8 = 0$$

$$k = 7 \quad C_9 = 2 * 2 * 2C_0/2 * 3 * 5 * 6 * 8 * 9$$

$$k = 8 \quad C_{10} = 2 * 2 * 2C_1/3 * 4 * 6 * 7 * 9 * 10$$

$$y = C_0 \left(1 + \frac{2}{2 * 3} x^3 + \frac{2 * 2}{2 * 3 * 5 * 6} x^6 + \frac{2 * 2 * 2}{2 * 3 * 5 * 6 * 8 * 9} x^9 + \dots \right) \\ + C_1 \left(x + \frac{2}{3 * 4} x^4 + \frac{2 * 2}{3 * 4 * 6 * 7} x^7 + \frac{2 * 2 * 2}{3 * 4 * 6 * 7 * 9 * 10} x^{10} + \dots \right)$$

$$31. \quad y' + 7xy = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 7x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 7 \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$\begin{matrix} k = n - 1 & k = n + 1 \\ n = k + 1 & n = k - 1 \end{matrix}$$

$$Si \quad n = 1 \rightarrow k = 0 \quad n = 0 \rightarrow k = 1$$

$$\sum_{k=0}^{\infty} (k+1) C_{k+1} x^k + 7 \sum_{k=3}^{\infty} C_{k-1} x^k = 0$$

$$C_1 = 0$$

$$\sum_{k=1}^{\infty} [(k+1) C_{k+1} + 7 C_{k-1}] x^k = 0$$

$$(k+1) C_{k+1} + 7 C_{k-1} = 0$$

$$C_{k+1} = \frac{-7 C_{k-1}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 1 \quad C_2 = -7C_0/2$$

$$k = 2 \quad C_3 = 0$$

$$k=3 \quad C_4 = 49C_0/8$$

$$k=4 \quad C_5 = 0$$

$$k=5 \quad C_6 = -343C_0/48$$

$$k=6 \quad C_7 = 0$$

$$k=7 \quad C_8 = 2401C_0/384$$

$$k=8 \quad C_9 = 0$$

$$y = C_0(1 - \frac{7}{2}x^2 + \frac{49}{8}x^4 - \frac{343}{48}x^6 + \frac{2401}{384}x^8 + \dots)$$

$$32. \quad y' + 2y = 0$$

$$\begin{aligned} \sum_{n=1}^{\infty} nC_n x^{n-1} + 2 \sum_{n=0}^{\infty} C_n x^n &= 0 \\ \sum_{n=0}^{\infty} (n+1)C_{n+1} x^n + 2 \sum_{n=0}^{\infty} C_n x^n &= 0 \\ \sum_{n=0}^{\infty} [(n+1)C_{n+1} + 2C_n] x^n &= 0 \\ (n+1)C_{n+1} + 2C_n &= 0 \\ C_{n+1} &= \frac{-2C_n}{n+1} \end{aligned}$$

Remplacemos k en la ecuación obtenida

$$n=0 \quad C_1 = -2C_0/1$$

$$n=1 \quad C_2 = -2C_1/2 = 2^2 C_1/2!$$

$$n=2 \quad C_3 = -2C_2/3 = 2^3 C_1/3!$$

$$C_n = (-1)^n \frac{2^n C_0}{n!}$$

$$y = \sum_{n=0}^{\infty} (-1)^n \frac{2^n C_0}{n!} x^n$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{(-2x)^n}{n!}$$

$$y = C_0 e^{-2x}$$

$$33. \quad y'' - 2xy' - 2y = 0$$

$$\sum_{n=0}^{\infty} n(n-1)C_n x^{n-2} - 2x \sum_{n=0}^{\infty} nC_n x^{n-1} - 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\begin{aligned}
& \sum_{n=0}^{\infty} n(n-1)C_n x^{n-2} - 2 \sum_{n=0}^{\infty} nC_n x^n - 2 \sum_{n=0}^{\infty} C_n x^n = 0 \\
& \sum_{n=0}^{\infty} (n+2)(n-1)C_n + 2x^n - 2 \sum_{n=0}^{\infty} nC_n x^n - 2 \sum_{n=0}^{\infty} C_n x^n = 0 \\
& (n+2)(n-1)C_{n+2} - 2nC_n - 2C_n = 0 \\
& C_{n+2} = \frac{2(n+1)C_n}{(n+2)(n+1)} = \frac{2C_n}{n+2}
\end{aligned}$$

Remplacemos k en la ecuación obtenida

$$n = 0 \quad C_2 = C_n$$

$$n = 1 \quad C_3 = 2C_1/3$$

$$n = 2 \quad C_4 = C_0/2$$

$$n = 3 \quad C_5 = 0$$

$$n = 4 \quad C_6 = C_0/6$$

$$y = C_0 \left(1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \dots \right) + C_1 \left(x + \frac{2x^3}{3} + \frac{2^2 x^5}{3 \cdot 5} + \dots \right)$$

$$34. \quad xy'' + (x+1)y = 1$$

$$\begin{aligned}
& x \sum_{n=2}^{\infty} (n-1)nC_n x^{n-2} + (x+1) \sum_{n=0}^{\infty} C_n x^n = 1 \\
& x \sum_{n=2}^{\infty} (n-1)nC_n x^{n-2} + x \sum_{n=0}^{\infty} C_n x^n + 1 \sum_{n=0}^{\infty} C_n x^n = 1 \\
& \sum_{n=2}^{\infty} (n-1)nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^{n+1} + \sum_{n=0}^{\infty} C_n x^n = 1 \\
& \quad \begin{matrix} k = n-1 & k = n+1 & k = n \\ n = k+1 & n = k-1 & k = 0 \end{matrix} \\
& \text{Si } n = 2 \rightarrow k = 1 \quad n = 0 \rightarrow k = 1 \\
& \sum_{k=1}^{\infty} k(k+1)C_{k+1}x^k + \sum_{k=1}^{\infty} C_{k-1}x^k + C_0 + \sum_{k=0}^{\infty} C_k x^k = 1 \\
& \quad C_0 = 0 \\
& \sum_{n=0}^{\infty} [k(k+1)C_{k+1} + C_{k-1} + C_k] x^n = 0 \\
& \quad (k+1)C_{k+1} + C_{k-1} + C_k = 0 \\
& \quad C_{k+1} = \frac{-C_{k-1} - C_k}{(k+1)k}
\end{aligned}$$

Remplacemos k en la ecuación obtenida

$$k = 1 \quad C_2 = -C_1/2 - 1/2$$

$$k = 2 \quad C_3 = -C_1/12 + 1/12$$

$$k = 3 \quad C_4 = 7/144 C_1 - 5/144$$

$$y = C_0 + C_1 x + \left(-\frac{C_1}{2} - \frac{1}{2}\right) x^2 + \left(-\frac{C_1}{12} + \frac{1}{12}\right) x^3 + \left(\frac{7}{144} C_1 - \frac{5}{144}\right) x^4 + \dots$$

$$y = C_0 + C_1 \left(x - \frac{1}{2} x^2 - \frac{1}{12} x^3 + \frac{7}{144} x^4 + \dots \right) - \frac{1}{2} x^2 + \frac{1}{12} x^3 - \frac{5}{144} x^4 + \dots$$

$$35. \quad y' - 4xy = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - 4x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - 4 \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$k = n - 1 \quad k = n + 1$$

$$n = k + 1 \quad n = k - 1$$

$$\text{Si } n = 1 \rightarrow k = 0 \quad n = 0 \rightarrow k = 1$$

$$C_1 + \sum_{k=0}^{\infty} (k+1) C_{k+1} x^k - 4 \sum_{k=1}^{\infty} C_{k-1} x^k = 0$$

$$C_1 = 0$$

$$\sum_{k=1}^{\infty} [(k+1) C_{k+1} - 4 C_{k-1}] x^k = 0$$

$$(k+1) C_{k+1} - 4 C_{k-1} = 0$$

$$C_{k+1} = \frac{4 C_{k-1}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 1 \quad C_2 = 2 C_0$$

$$k = 2 \quad C_3 = 0$$

$$k = 3 \quad C_4 = 2 C_0$$

$$k = 4 \quad C_5 = 0$$

$$k = 5 \quad C_6 = 4 C_0 / 3$$

$$k = 6 \quad C_7 = 0$$

$$k = 7 \quad C_8 = 2 C_0 / 3$$

$$k = 8 \quad C_9 = 0$$

$$y = C_0 \left(1 + 2x^2 + 2x^4 + \frac{4}{3} x^6 + \frac{2}{3} x^8 + \dots \right)$$

36. $y'' - xy' + 2y = 0$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - \sum_{n=1}^{\infty} nC_n x^n + 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{k=0}^{\infty} (k+2)(k+1)C_k + 2x^k - \sum_{k=1}^{\infty} kC_k x^k + 2 \sum_{k=0}^{\infty} C_k x^k$$

$$2C_2 + 2C_0 + \sum_{k=0}^{\infty} [(k+2)(k+1)C_k + 2 - (k-2)C_k]x^k = 0$$

$$2C_2 + 2C_0 = 0$$

$$(k+2)(k+1)C_k + 2 - (k-2)C_k = 0$$

$$C_2 = -C_0$$

CASO 1, $C_0 = 1$ y $C_1 = 0$

$$C_2 = -1$$

$$C_3 = C_5 = C_7 = 0$$

$$C_4 = 0$$

$$C_6 = C_8 = C_{10} = 0$$

CASO 2, $C_0 = 0$ y $C_1 = 1$

$$C_2 = C_4 = C_6 = 0$$

$$C_3 = -1/6$$

$$C_5 = -1/120$$

$$y_1 = 1 - x^2$$

$$y_2 = x - \frac{1}{6}x^3 - \frac{1}{120}x^5 - \dots$$

37. $y'' + x^2y' + xy = 0$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$\sum_{k=0}^{\infty} (k+2)(k+1)C_k + 2x^k + \sum_{k=2}^{\infty} (k+1)C_k - 1x^k + \sum_{k=1}^{\infty} C_{k-1}x^n$$

$$2C_2 + (6C_3 + C_0)x + \sum_{k=2}^{\infty} [(k+2)(k+1)C_{k+2} + kC_{k-1}]x^k = 0$$

$$C_2 = 6C_3 + C_0 = 0$$

$$(k+2)(k+1)C_k + kC_{k-1} = 0$$

$$C_3 = -1/6 C_0$$

$$C_{k+2} = \frac{k}{(k+2)(k+1)} C_{k-1}$$

CASO 1, $C_0 = 1$ y $C_1 = 0$

$$C_2 = -1/6 C_0$$

$$C_4 = C_5 = 0$$

$$C_6 = -1/45$$

CASO 2, $C_0 = 0$ y $C_1 = 1$

$$C_3 = 0$$

$$C_4 = -1/6$$

$$C_5 = C_6 = 0$$

$$C_7 = 5/252$$

$$y_1 = 1 - \frac{1}{6}x^3 + \frac{1}{45}x^6 - \dots$$

$$y_2 = x - \frac{1}{6}x^4 + \frac{5}{232}x^7 - \dots$$

$$38. y' + y = 0$$

$$\sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$k = n - 1 \quad k = n$$

$$\sum_{k=0}^{\infty} (k+1)C_{k+1}x^k + \sum_{k=0}^{\infty} C_k x^k = 0$$

$$\sum_{k=0}^{\infty} [(k+1)C_{k+1} + C_k]x^k = 0$$

$$(k+1)C_{k+1} + C_k = 0$$

$$C_{k+1} = -\frac{C_k}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 0 \quad C_1 = -C_0$$

$$k = 1 \quad C_2 = C_0/2$$

$$k = 2 \quad C_3 = -C_0/6$$

$$k = 3 \quad C_4 = C_0/24$$

$$k = 4 \quad C_5 = -C_0/120$$

$$y = C_0(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!})$$

$$39. y' - 2y = 0$$

$$\sum_{n=1}^{\infty} nC_n x^{n-1} - 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$k = n - 1 \quad k = n$$

$$\sum_{k=0}^{\infty} (k+1)C_{k+1}x^k - 2 \sum_{k=0}^{\infty} C_k x^k = 0$$

$$\sum_{k=0}^{\infty} [(k+1)C_{k+1} - 2C_k]x^k = 0$$

$$(k+1)C_{k+1} - 2C_k = 0$$

$$C_{k+1} = \frac{2C_k}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 0 \quad C_1 = 2C_0$$

$$\begin{aligned}
k=1 & \quad C_2 = 2C_0 \\
k=2 & \quad C_3 = 4C_0/3 \\
k=3 & \quad C_4 = 8C_0/12 \\
k=4 & \quad C_5 = 16C_0/60
\end{aligned}$$

$$y = C_0(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{8}{12}x^4 + \frac{16}{60}x^5)$$

40. $y' - x^2y = 0$

$$\begin{aligned}
\sum_{n=1}^{\infty} nC_n x^{n-1} - x^2 \sum_{n=0}^{\infty} C_n x^n &= 0 \\
\sum_{k=0}^{\infty} nC_n x^{n-1} - \sum_{k=0}^{\infty} C_k x^{k+2} &= 0 \\
C_1 + C_2 x + \sum_{n=3}^{\infty} nC_n x^{n-1} - \sum_{n=0}^{\infty} C_n x^{n+2} &= 0 \\
C_1 &= 0 \\
C_2 &= 0 \\
\sum_{k=0}^{\infty} (k+3)C_{k+3} x^{k+2} - \sum_{n=0}^{\infty} C_n x^{n+2} &= 0 \\
\sum_{k=0}^{\infty} [(k+3)C_{k+3} - C_k] x^{k+2} &= 0 \\
(k+3)C_{k+3} - C_k &= 0 \\
C_{k+3} &= \frac{C_k}{k+3}
\end{aligned}$$

Remplacemos k en la ecuación obtenida

$$\begin{aligned}
k=0 & \quad C_3 = C_0/3 \\
k=1 & \quad C_4 = 0 \\
k=2 & \quad C_5 = 0 \\
k=3 & \quad C_6 = C_0/18 \\
k=4 & \quad C_7 = 0 \\
k=5 & \quad C_8 = 0 \\
k=6 & \quad C_9 = C_0/162
\end{aligned}$$

$$y = C_0(1 + \frac{x^3}{3} + \frac{x^6}{18} + \frac{x^9}{162})$$

41. $y'' + y' - 2y = 0 \quad y(0) = 1, y'(0) = 2$

$$\begin{aligned}
\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} nC_n x^{n-1} - 2 \sum_{n=0}^{\infty} C_n x^n &= 0 \\
\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} nC_n x^{n-1} - \sum_{n=0}^{\infty} 2C_n x^n &= 0
\end{aligned}$$

$$\sum_{n=2}^{\infty} (n+2)(n-1)C_{n+2}x^n + \sum_{n=0}^{\infty} (n+1)C_{n+1}x^n - \sum_{n=0}^{\infty} 2C_nx^n = 0$$

$$\sum_{k=0}^{\infty} [(n+2)(n-1)C_{n+2} + (n+1)C_{n+1} - 2C_n]x^n = 0$$

$$(n+2)(n-1)C_{n+2} + (n+1)C_{n+1} - 2C_n = 0$$

$$(n+2)(n-1)C_{n+2} = -(n+1)C_{n+1} + 2C_n$$

$$C_{n+2} = \frac{2C_n - (n+1)C_{n+1}}{(n+2)(n+1)}$$

Remplacemos k en la ecuación obtenida

$$n=0 \quad C_2 = 0$$

$$n=1 \quad C_3 = -4/3$$

$$n=2 \quad C_4 = 2/3$$

$$n=3 \quad C_5 = -4/15$$

$$n=4 \quad C_6 = 4/45$$

$$n=5 \quad C_7 = -8/315$$

$$n=6 \quad C_8 = 2/315$$

$$y = C_0(1 + \frac{x^3}{3} + \frac{x^6}{18} + \frac{x^9}{162})$$

$$42. \quad y' + 3y = 0$$

$$\sum_{n=1}^{\infty} nC_nx^{n-1} + 3 \sum_{n=0}^{\infty} C_nx^n = 0$$

$$\sum_{k=0}^{\infty} (k+1)C_{k+1}x^k + 3 \sum_{k=0}^{\infty} C_kx^k = 0$$

$$\sum_{k=0}^{\infty} [(k+1)C_{k+1} + 3C_k]x^k = 0$$

$$(k+1)C_{k+1} + 3C_k = 0$$

$$C_{k+1} = \frac{-3C_k}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k=0 \quad C_1 = -3C_0$$

$$k=1 \quad C_2 = 9/2 C_0$$

$$k=2 \quad C_3 = -9/2 C_0$$

$$k=3 \quad C_4 = 27/8 C_0$$

$$y = C_0(1 - 3x + \frac{9}{2}x^2 - \frac{9}{2}x^3 + \frac{27}{8}x^4 + \dots)$$

$$43. \quad y' + 7xy = 0$$

$$\sum_{n=1}^{\infty} nC_nx^{n-1} + 7x \sum_{n=0}^{\infty} C_nx^n = 0$$

$$\sum_{n=1}^{\infty} nC_nx^{n-1} + 7 \sum_{n=0}^{\infty} C_nx^{n+1} = 0$$

$$\sum_{k=0}^{\infty} (k+1)C_{k+1}x^k + 7 \sum_{k=1}^{\infty} C_{k-1}x^k = 0$$

$$C_1 + \sum_{k=1}^{\infty} (k+1)C_{k+1}x^k + 7 \sum_{k=1}^{\infty} C_{k-1}x^k = 0$$

$$C_1 = 0$$

$$\sum_{k=0}^{\infty} [(k+1)C_{k+1} + 7C_{k-1}]x^k = 0$$

$$(k+1)C_{k+1} + 7C_{k-1} = 0$$

$$C_{k+1} = \frac{-7C_{k-1}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k=1 \quad C_2 = -\frac{7C_0}{2}$$

$$k=2 \quad C_3 = 0$$

$$k=3 \quad C_4 = \frac{49C_0}{8}$$

$$k=4 \quad C_5 = 0$$

$$k=5 \quad C_6 = -\frac{343C_0}{48}$$

$$k=6 \quad C_7 = 0$$

$$k=7 \quad C_8 = \frac{2401C_0}{384}$$

$$k=8 \quad C_9 = 0$$

$$y = C_0 \left(1 - \frac{7}{2}x^2 + \frac{49}{8}x^4 - \frac{343}{48}x^6 + \frac{2401}{384}x^8 \dots \right)$$